

Computation Graphs, Backpropagation and Autodiff

How gradients flow through neural networks

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ES 667 · Deep Learning · IIT Gandhinagar

Backprop vocabulary

Lecture 2 built the forward map; today we reverse it

Last time, an MLP turned an input into a prediction and then one scalar loss. Today we reuse that **same graph** in reverse:

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General pattern

Forward: compute values

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Backward: compute sensitivities loss
sensitivity for every parameter θ_j

Concrete scalar example

Forward: with $(w, x, b, y) = (2, 3, 1, 10)$

$$\hat{y} = 2 \cdot 3 + 1 = 7, \quad \mathcal{L} = (7 - 10)^2 = 9$$

Backward:

$$\left(\frac{\partial \mathcal{L}}{\partial w}, \frac{\partial \mathcal{L}}{\partial b} \right) = (-18, -6)$$

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same executed graph · values flow forward, gradients flow backward

Read the notation before the algorithm

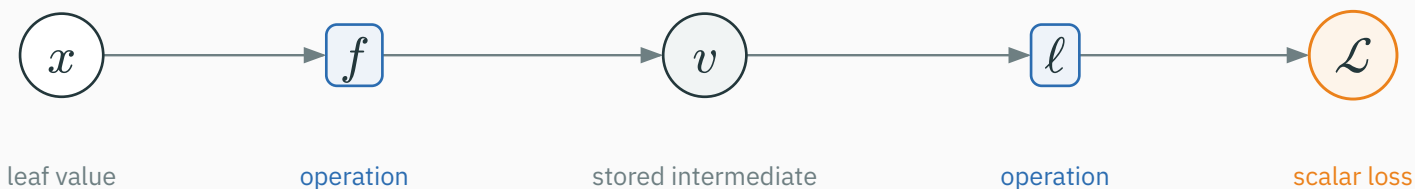
Symbol	Meaning in this lecture
u, v, z	scalars
$\mathbf{x}, \mathbf{z}, \boldsymbol{\theta}$	vectors (bold lowercase)
$\mathbf{W}, \mathbf{X}$	matrices (bold uppercase)
$\partial \mathcal{L} / \partial u$	how the loss changes when scalar u changes
$\partial \mathcal{L} / \partial \boldsymbol{\theta}$	all parameter derivatives, arranged like $\boldsymbol{\theta}$

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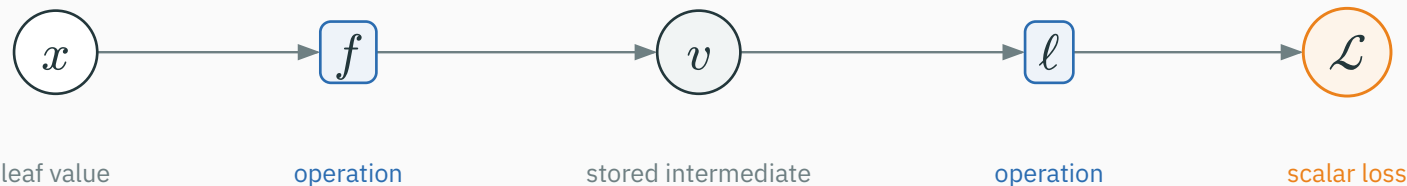
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We keep the full $\partial\mathcal{L}/\partial(\cdot)$ notation visible throughout: the numerator is always the final loss; the denominator names the stored value whose derivative we want.

Our computation-graph convention

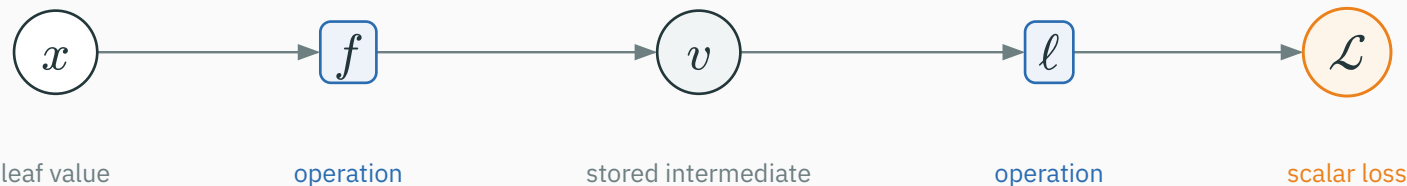


Our computation-graph convention



white circle	given value: input, parameter, or target — a graph leaf
gray circle	computed intermediate value, saved for backward when needed
blue box	operation: consumes value(s) and produces a value
gray arrow	forward dependency from a value to the operation that uses it
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circles store values • boxes transform them • arrows show dependency

Use the same scalar example throughout:

$$\mathcal{L} = (wx + b - y)^2, \quad (w, x, b, y) = (2, 3, 1, 10).$$

One derivative, three routes

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Route	Computation	For this example
symbolic	differentiate the whole expression	$\frac{\partial \mathcal{L}}{\partial w} = 2(wx + b - y)x = -18$
finite difference	probe $w \pm \varepsilon$	$\frac{\mathcal{L}(2.01) - \mathcal{L}(1.99)}{2\varepsilon} \approx -18$

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Same derivative; different computation.

Autodiff gives an analytic gradient; central difference gives an independent numerical estimate: $g_{\text{FD}} = \frac{\mathcal{L}(\theta_j + \varepsilon) - \mathcal{L}(\theta_j - \varepsilon)}{2\varepsilon}$.

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import torch
def loss(w): return (w*3.0 + 1.0 - 10.0)**2
w, eps = 2.0, 1e-5
g_fd = (loss(w + eps) - loss(w - eps)) / (2*eps)
wt = torch.tensor(w, dtype=torch.float64, requires_grad=True)
loss(wt).backward()
g_ad = wt.grad.item()
print(g_fd, g_ad) # both approximately -18
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Use float64 and check only a few coordinates. Finite differences debug gradients; they do not train models because every checked coordinate needs two extra forward passes.

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Symbolic to reason · finite differences to check · reverse mode to train.

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We need all P derivatives $\partial \mathcal{L} / \partial \theta_j$. Backprop computes them together in **one reverse sweep**, without perturbing parameters one at a time.

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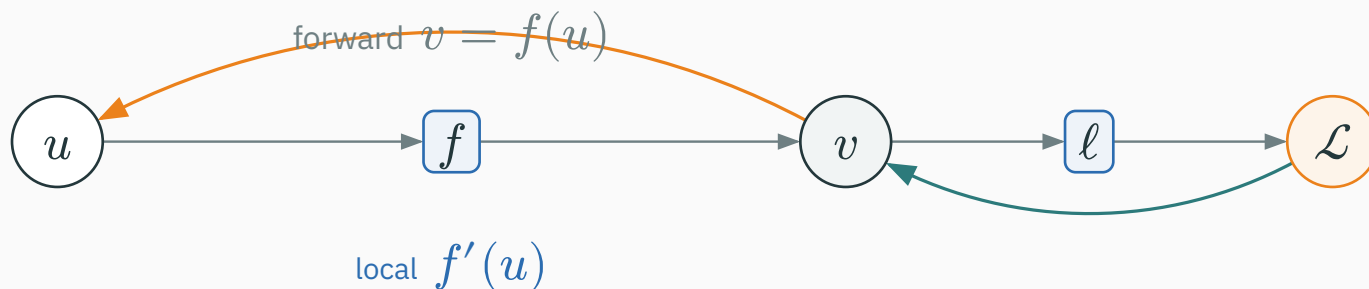
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5 · depth + gradient flow	What happens when many local gradients multiply?

One local operation

Zoom in on one operation. The value circle u enters the boxed operation f ; its output is stored in the value circle v .

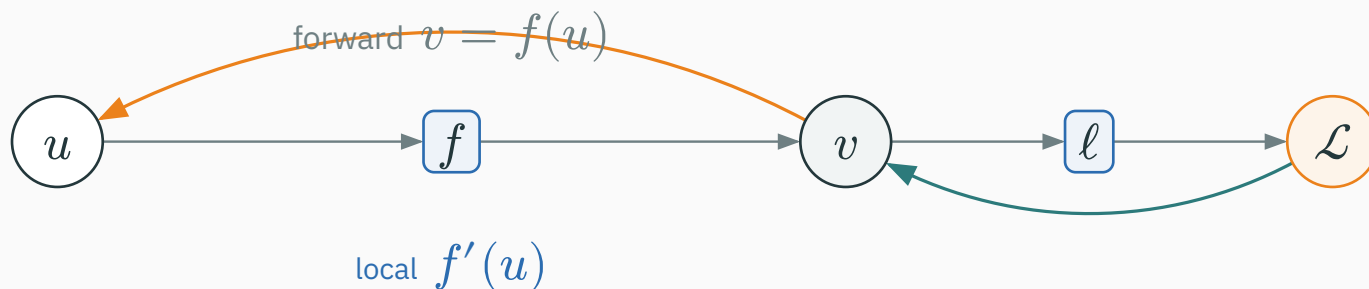
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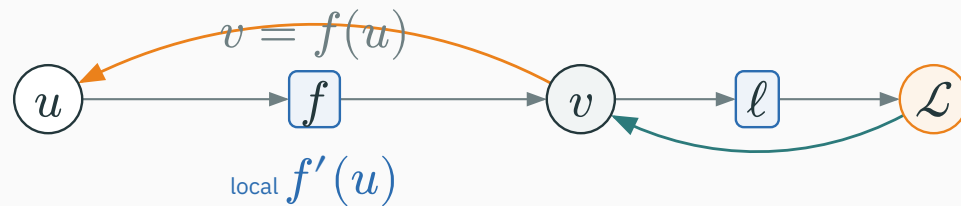


backward computes $\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial v} \cdot f'(u)$

Every operation multiplies the arriving gradient by its local gradient

D DERIVATION

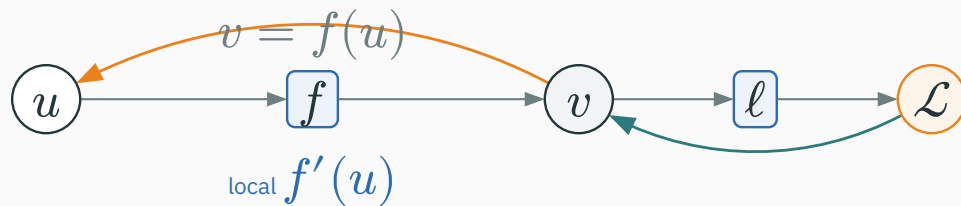
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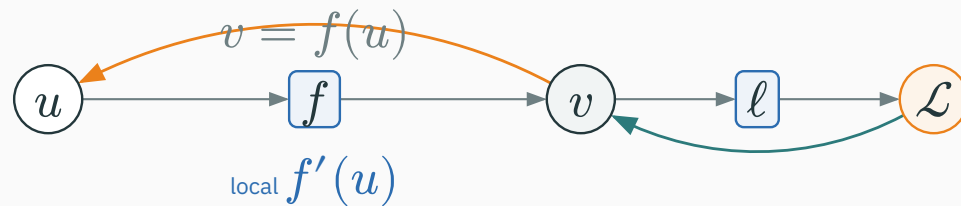


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Downstream gradient

sent back to u

$$\frac{\partial \mathcal{L}}{\partial u}$$

Upstream gradient

arriving at v

$$\frac{\partial \mathcal{L}}{\partial v}$$

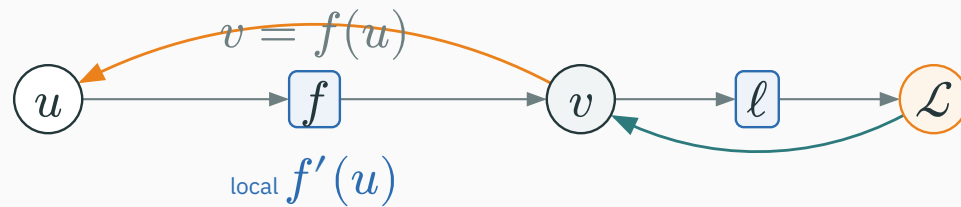
Local gradient

this operation's slope

$$f'(u) = \frac{\partial v}{\partial u}$$

The chain rule is one reusable local rule

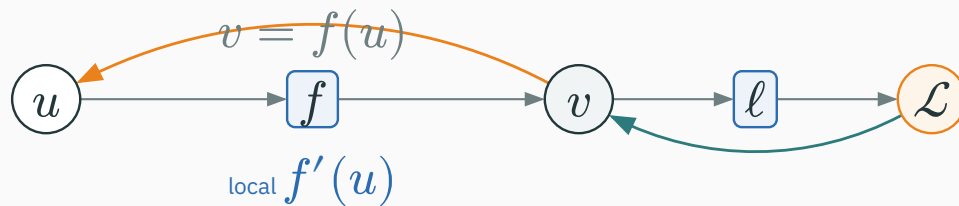
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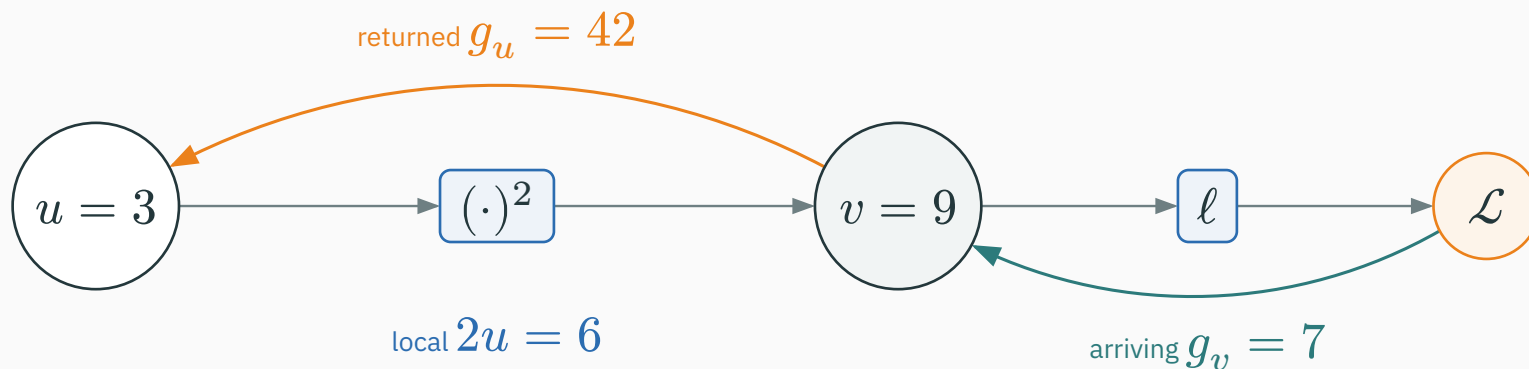
The operation multiplies its **upstream gradient** by its **local gradient** to produce the **downstream gradient**. It needs no global formula for the whole network.

Example: a square operation

For $v = u^2$ at $u = 3$, the forward value is $v = 9$ and the **local slope** is $\partial v / \partial u = 2u = 6$. Suppose the rest of the graph sends $g_v = \partial \mathcal{L} / \partial v = 7$ back to this square.

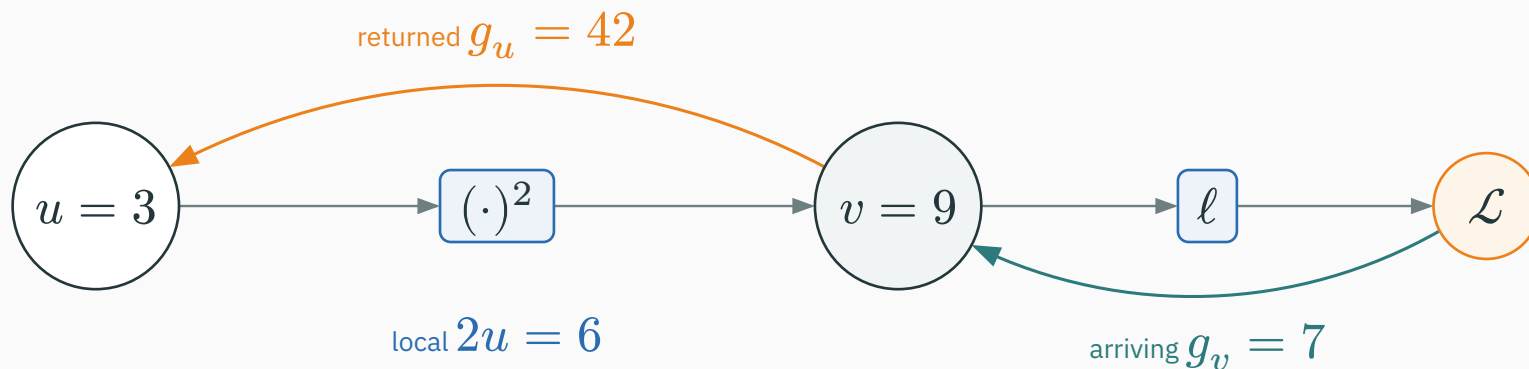
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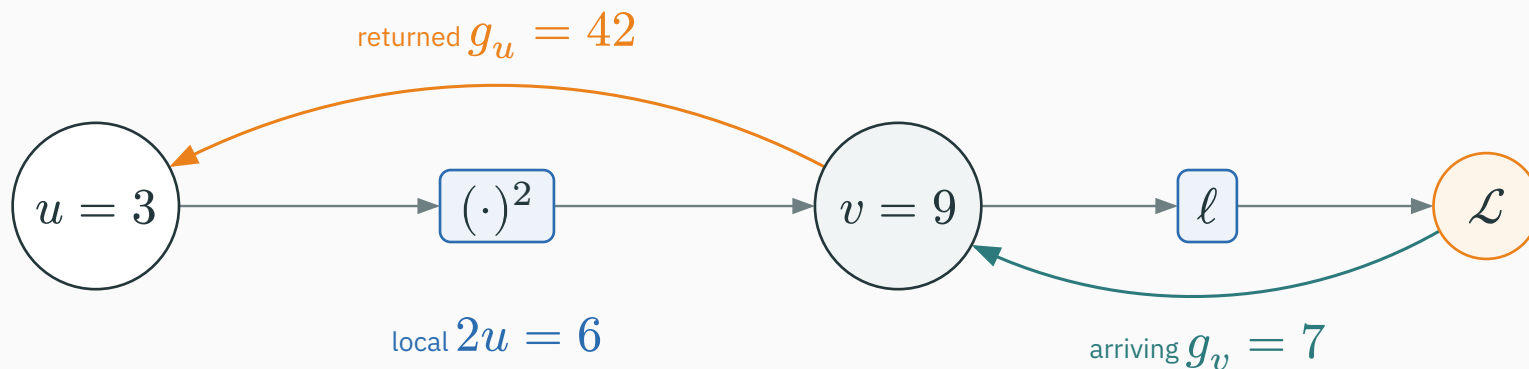
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$$\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial v} \cdot 2u = 7 \cdot 6 = 42.$$

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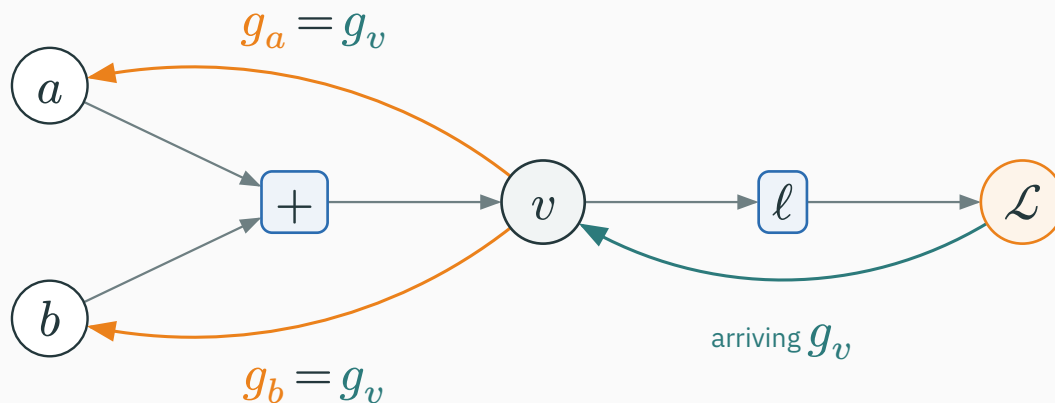
The square operation multiplies the arriving $\partial \mathcal{L} / \partial v = 7$ by its **local gradient** $2u = 6$ to produce $\partial \mathcal{L} / \partial u = 42$.

Addition copies the arriving gradient

Operation $v = a + b$. Let $g_v = \partial \mathcal{L} / \partial v$ arrive. Local derivatives: $\partial v / \partial a = 1$, $\partial v / \partial b = 1$.

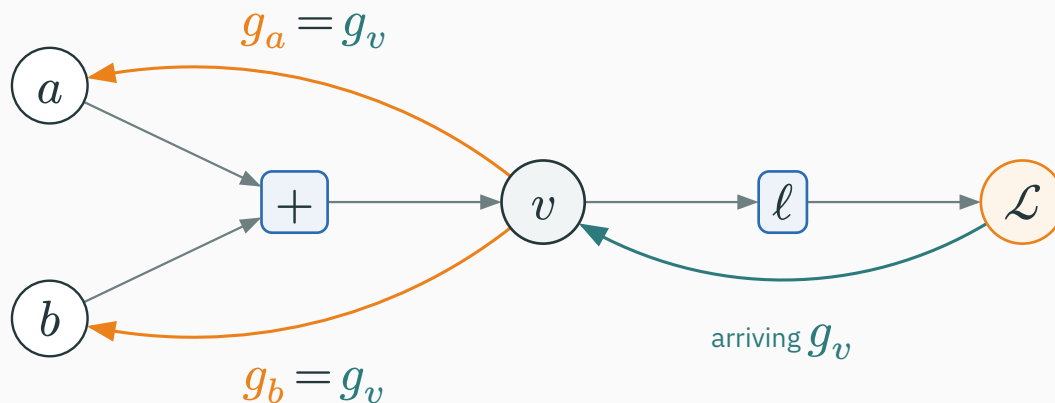
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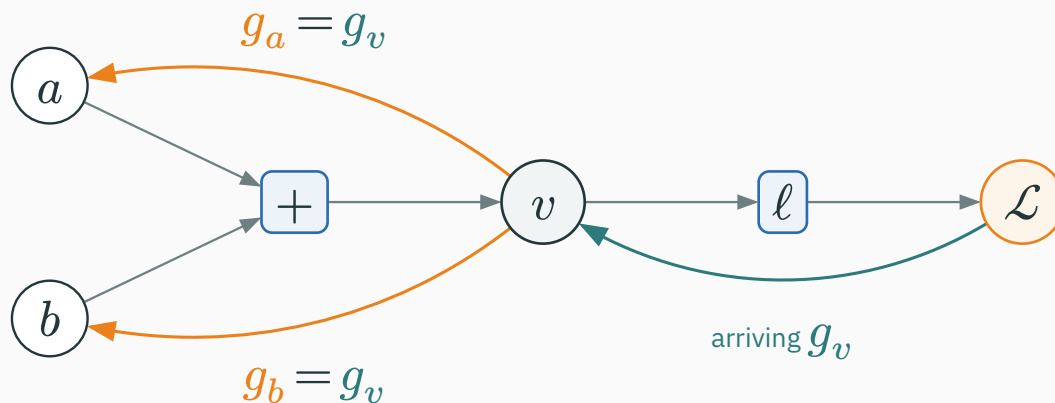
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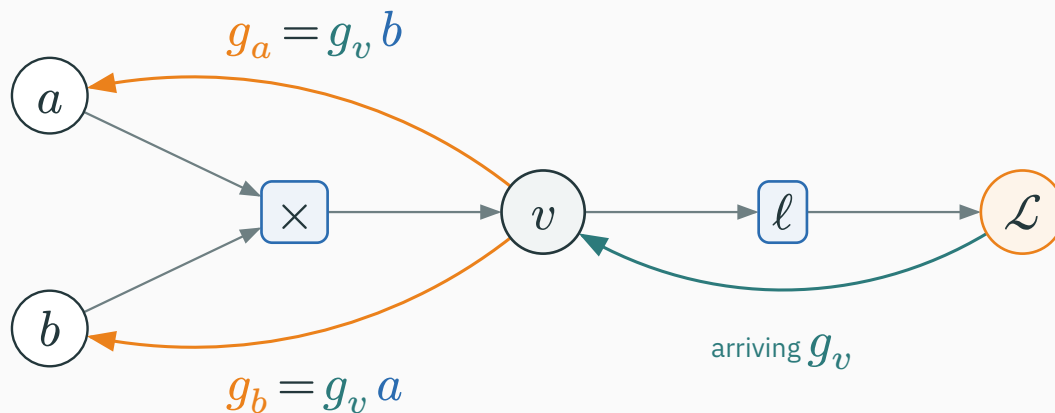
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+ copies the arriving derivative into a downstream derivative for every input.

Operation $v = ab$. Let $g_v = \partial \mathcal{L} / \partial v$ arrive. Local derivatives: $\partial v / \partial a = b$, $\partial v / \partial b = a$.

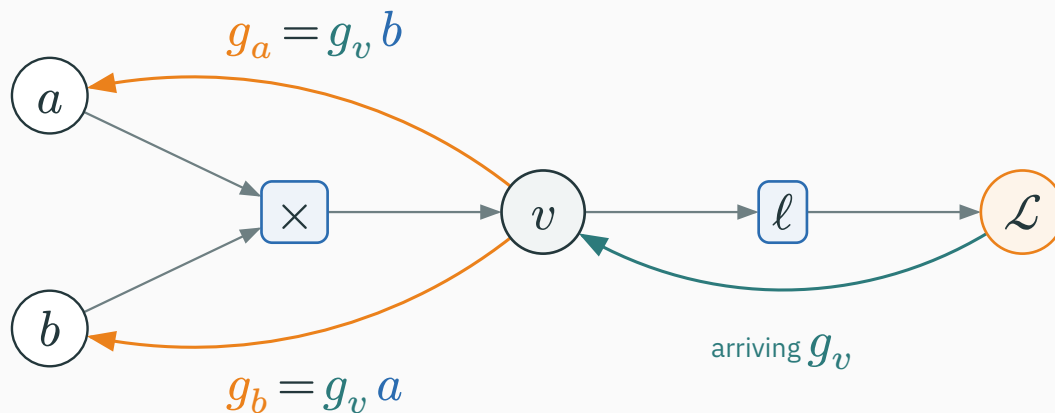
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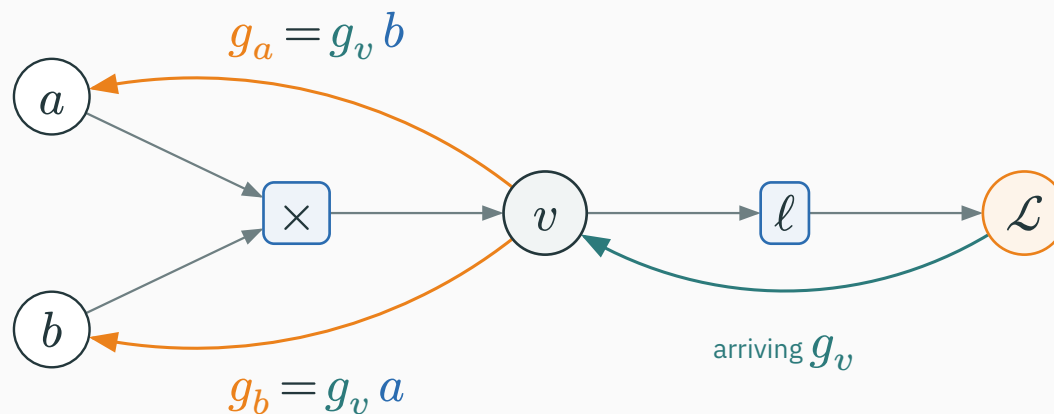
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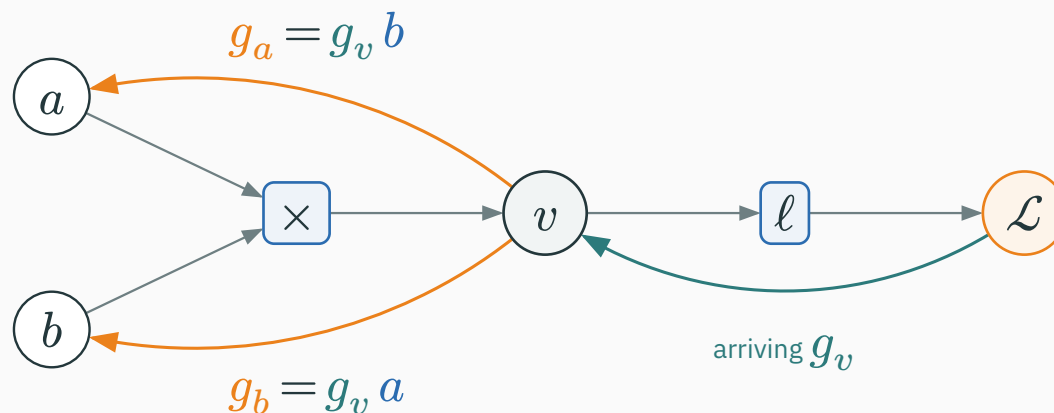


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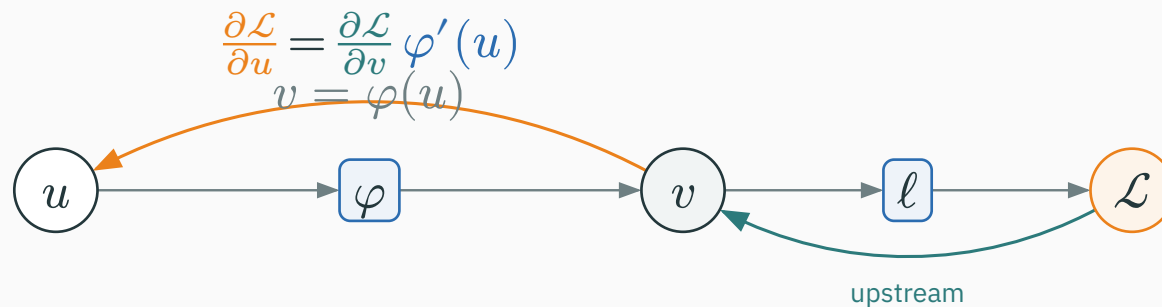
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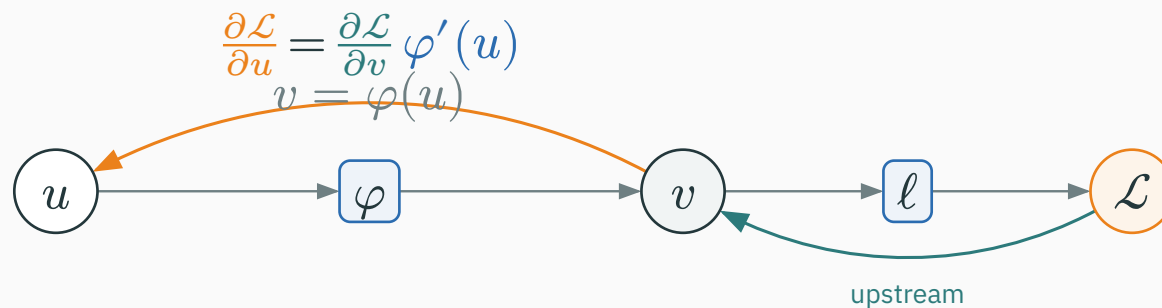
| If $a = 2$, $b = 5$, and $\partial \mathcal{L} / \partial v = 3$, then $\partial \mathcal{L} / \partial a = 15$ and $\partial \mathcal{L} / \partial b = 6$.

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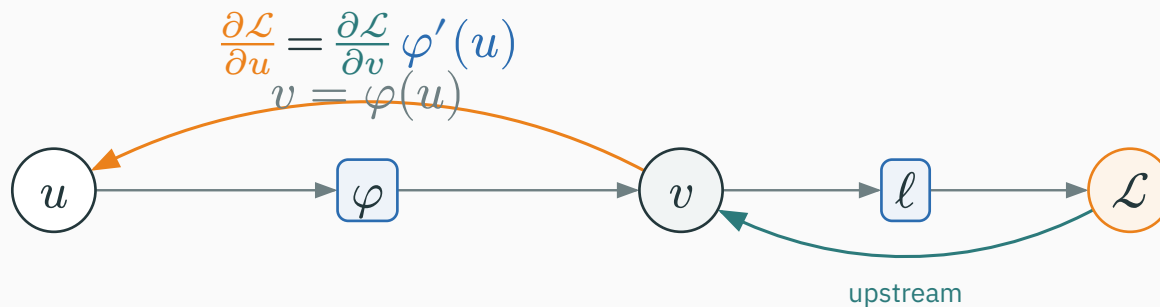


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| **sigmoid** σ : $\varphi'(u) = \sigma(u)(1 - \sigma(u))$

| **ReLU**: $\varphi'(u) = 1$ for $u > 0$, and 0 for $u < 0$

Pop quiz: what crosses an inactive ReLU?

QUESTION

Let $v = \text{ReLU}(u)$. If $u = -2$ and the upstream gradient is $\partial \mathcal{L} / \partial v = 5$, what downstream gradient $\partial \mathcal{L} / \partial u$ is sent back?

A. 5

B. -10

C. 0

D. undefined because $u < 0$

Answer: an inactive ReLU blocks the gradient

A ANSWER

Correct: C — 0

Why: For $u < 0$, the local gradient is $\partial v / \partial u = 0$. Therefore downstream = upstream $\times$ local = $5 \times 0 = 0$.

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| **Backward:** $5 \times 0 \rightarrow \partial \mathcal{L} / \partial u = 0$

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Backward: $5 \times 0 \rightarrow \partial \mathcal{L} / \partial u = 0$

At exactly $u = 0$, ReLU's ordinary derivative is undefined. Autodiff software must choose a convention; commonly it returns 0.

The rule table — memorize this

Forward operation	Contribution sent backward
$v = a + b$	$\frac{\partial \mathcal{L}}{\partial a} += \frac{\partial \mathcal{L}}{\partial v} \cdot 1$ $\frac{\partial \mathcal{L}}{\partial b} += \frac{\partial \mathcal{L}}{\partial v} \cdot 1$
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$v = \varphi(u)$	$\frac{\partial \mathcal{L}}{\partial u} += \frac{\partial \mathcal{L}}{\partial v} \cdot \varphi'(u)$

Use +=, not = — a value consumed by several operations accumulates gradient from **all** of them.

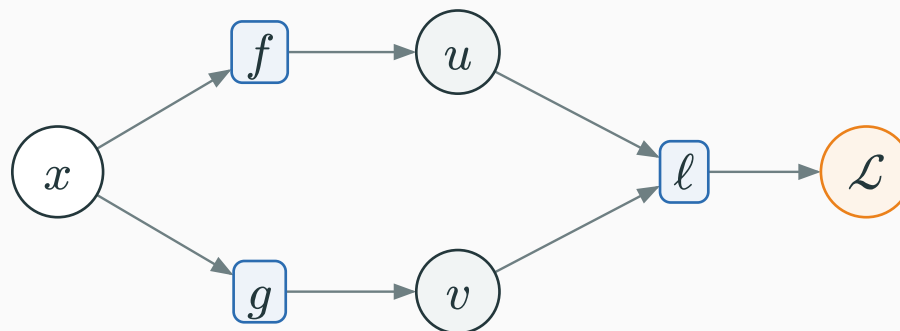
Gradients accumulate at branches

Why gradients add at a branch

Imagine nudging x from 4 to 4.01. The **same nudge** travels down both routes to the loss:

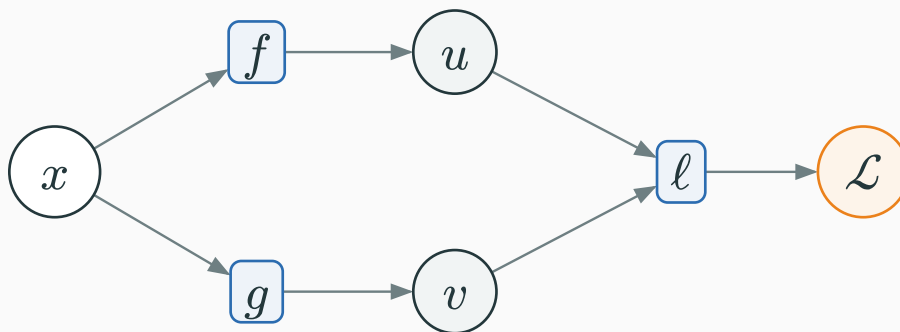
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Why gradients add at a branch

Imagine nudging x from 4 to 4.01. The **same nudge** travels down both routes to the loss:



Via $u = x^2$: the local slope is $2x = 8$, so $\Delta x = 0.01$ changes the loss by about $8 \cdot 0.01 = 0.08$.

Via $v = 3x$: the local slope is 3, so the same nudge changes the loss by about $3 \cdot 0.01 = 0.03$.

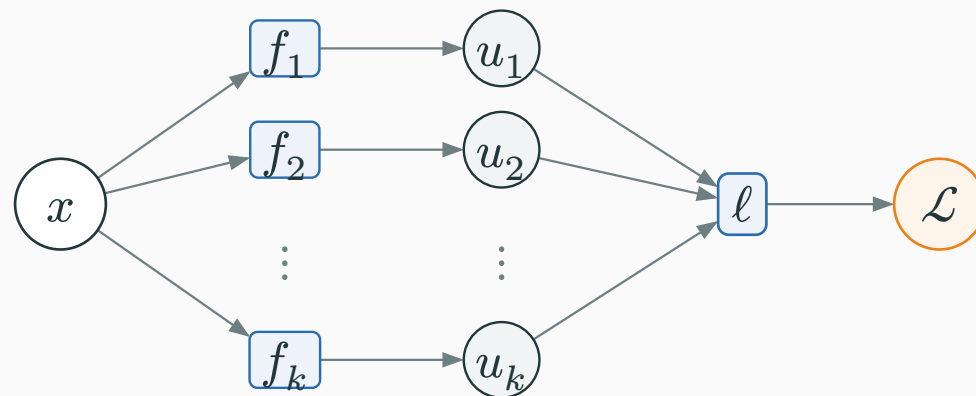
$$\text{total slope} = \text{total loss change} \div \text{nudge} = \frac{0.08+0.03}{0.01} = 8 + 3 = 11$$

The multivariable chain rule adds every path

The nudge experiment generalizes. If x influences $\mathcal{L}$ through $u_1, \dots, u_k$:

The multivariable chain rule adds every path

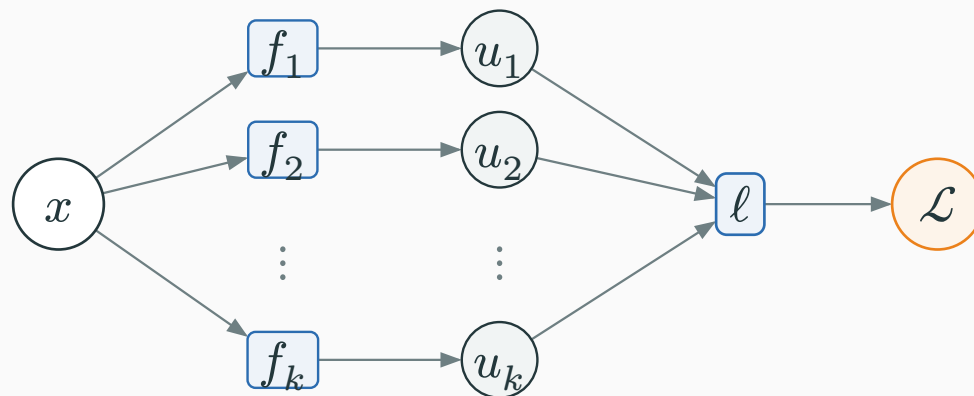
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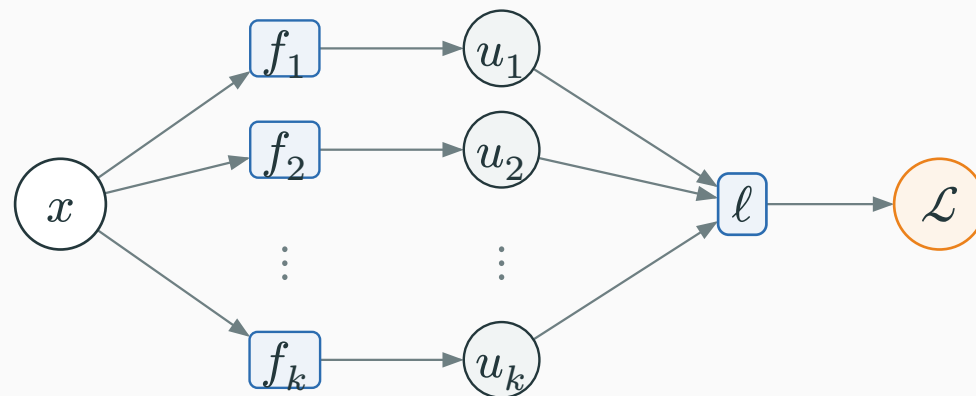


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each path contributes one chain-rule product; the products add

Worked branch example

Let $u = x^2$, $v = 3x$, $\mathcal{L} = u + v = x^2 + 3x$.

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Now let us get the **same** answer by backprop.

Forward at $x = 4$: $u = 16$, $v = 12$, $\mathcal{L} = 28$.

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$$\frac{\partial\mathcal{L}}{\partial x} = 8 + 3 = 11. \quad \checkmark$$

PyTorch adds the same two paths

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import torch

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v = 3*x          # path 2
L = u + v
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autograd accumulates both downstream contributions: $8 + 3 = 11$

The central worked example

Predict $\hat{y} = wx + b$, squared-error loss $\mathcal{L} = (\hat{y} - y)^2$.

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Compute $\partial\mathcal{L}/\partial w$, $\partial\mathcal{L}/\partial x$, $\partial\mathcal{L}/\partial b$, and $\partial\mathcal{L}/\partial y$ by backpropagation; verify the four values with PyTorch.

Break into primitives

Decompose the loss into elementary operations and name each stored result:

$$m = wx, \quad a = m + b, \quad e = a - y, \quad \mathcal{L} = e^2.$$

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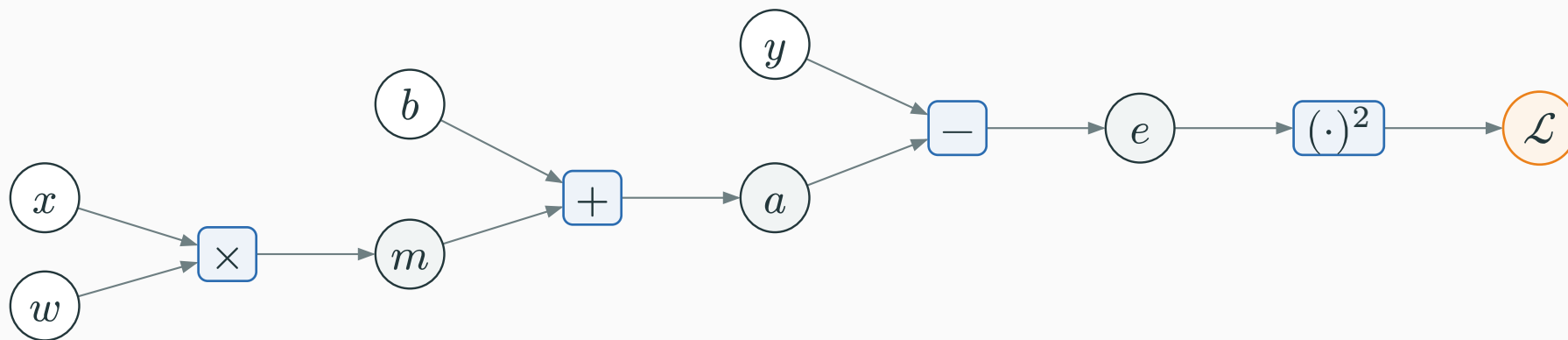
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value m after $\times$ • value a after $+$ • value e after $-$ • value $\mathcal{L}$ after $(\cdot)^2$

The symbols m , a , and e name stored intermediate values. They are circles in the graph; the four operations are boxes.

The computation graph



Given values	w, x, b, y	Stored values	m, a, e
Operations	$\times, +, -, (\cdot)^2$	Scalar loss	$\mathcal{L}$

Stored value	Operation that produces it	Numeric value
m	$w \times x$	$2 \cdot 3 = 6$
a	$m + b$	$6 + 1 = 7$
e	$a - y$	$7 - 10 = -3$
$\mathcal{L}$	e^2	$(-3)^2 = 9$

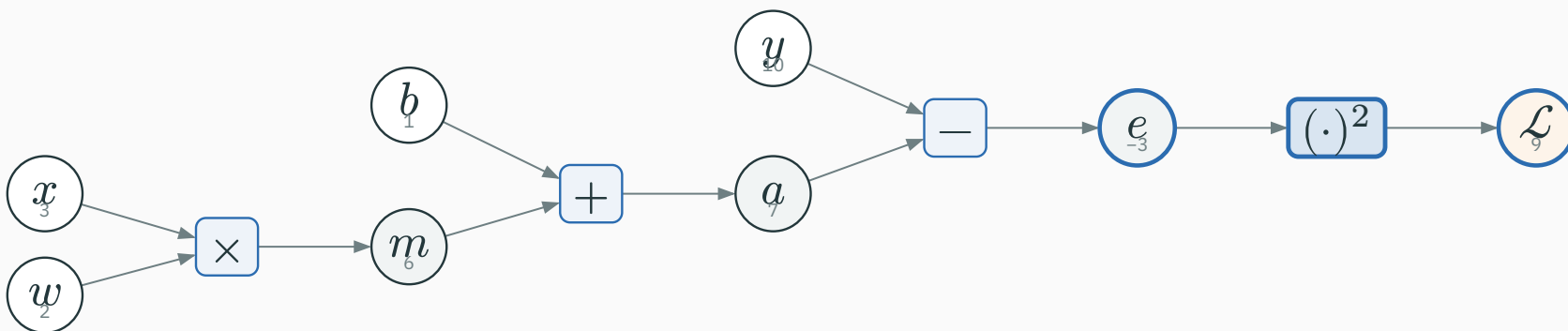
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$\mathcal{L} = 9$ – now run the graph **backward**

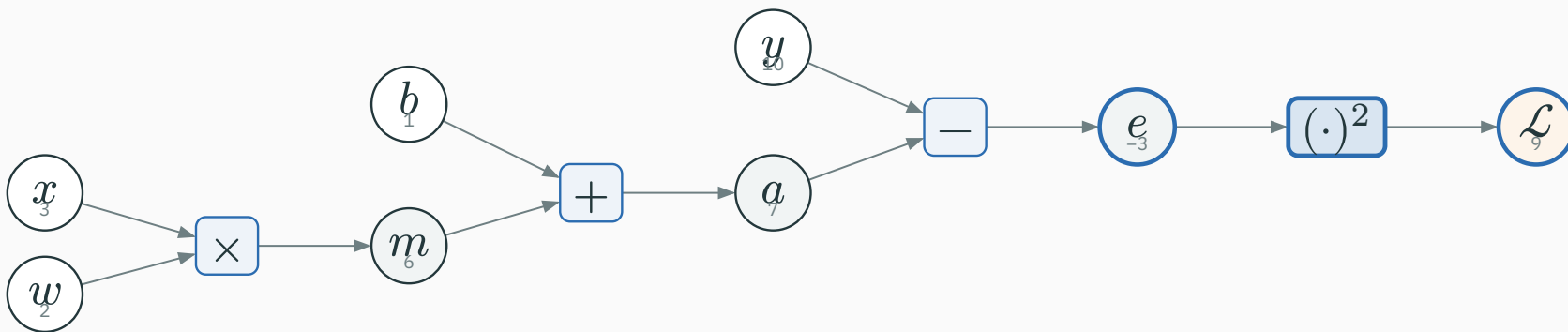
Seed the output: $\partial \mathcal{L} / \partial \mathcal{L} = 1$.

Backward: seed and the square operation

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Square operation $\mathcal{L} = e^2$, local $\partial \mathcal{L} / \partial e = 2e$:

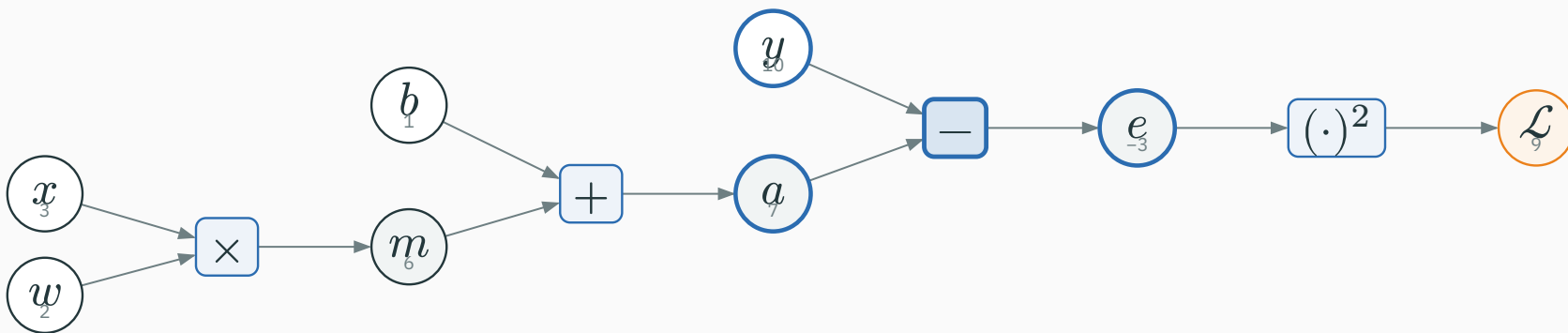
$$\frac{\partial \mathcal{L}}{\partial e} = 1 \cdot 2e = 1 \cdot 2(-3) = -6.$$

Backward: the subtraction operation

Subtraction operation $e = a - y$, locals $\partial e / \partial a = 1$, $\partial e / \partial y = -1$:

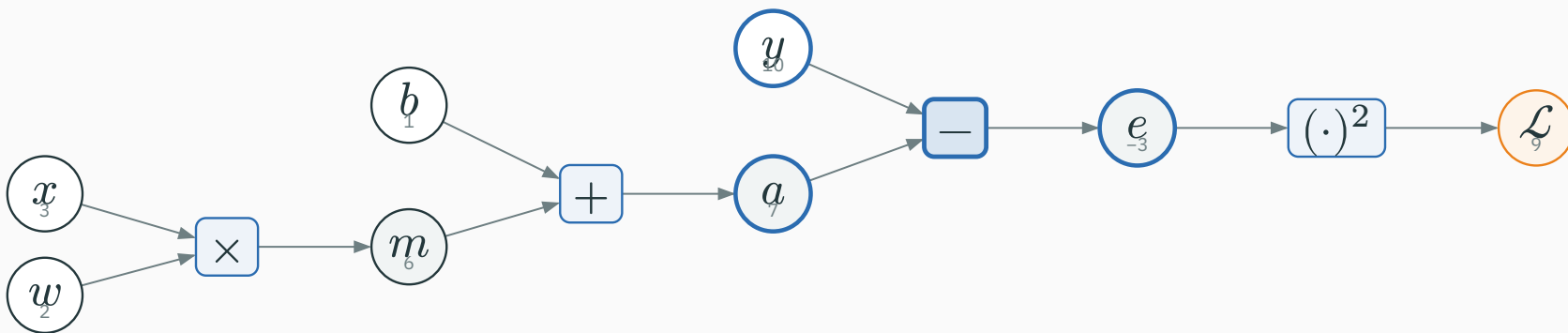
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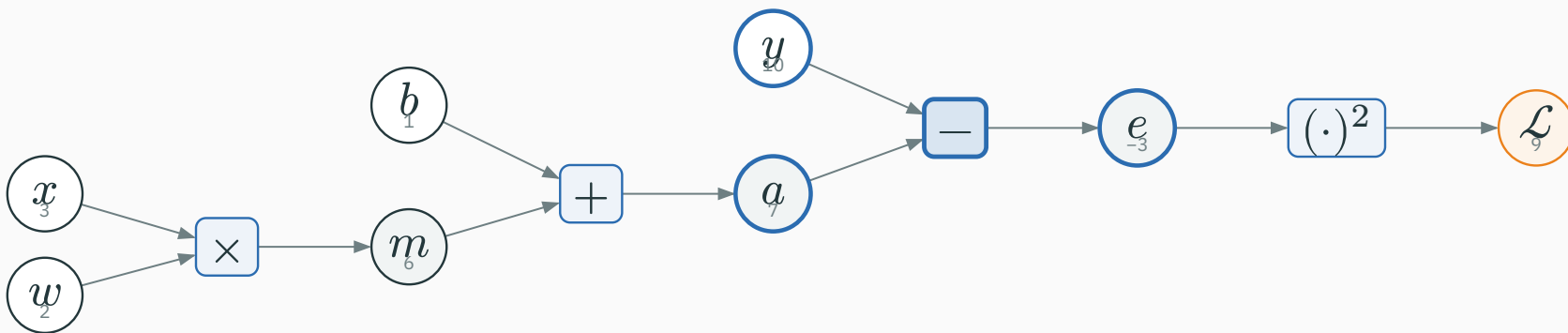
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$$\frac{\partial \mathcal{L}}{\partial a} = \frac{\partial \mathcal{L}}{\partial e} \cdot 1 = -6$$

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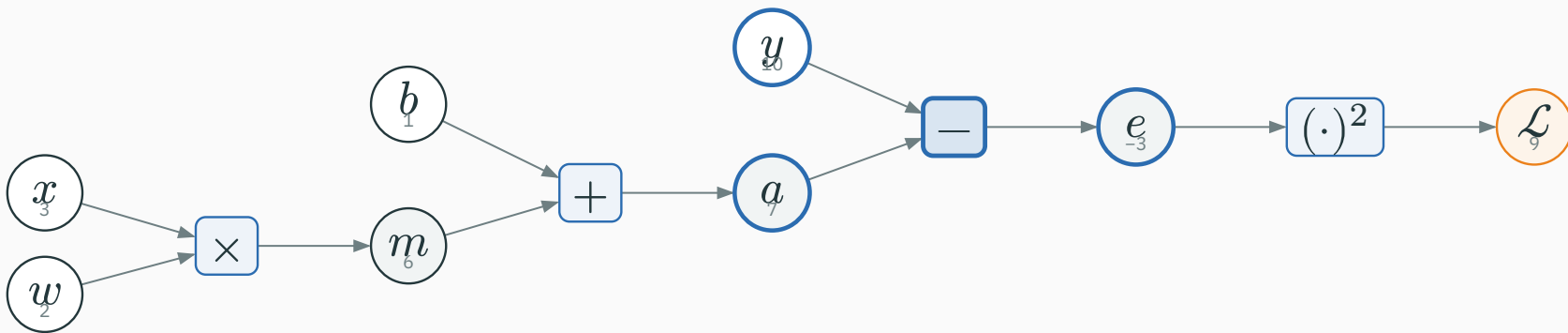
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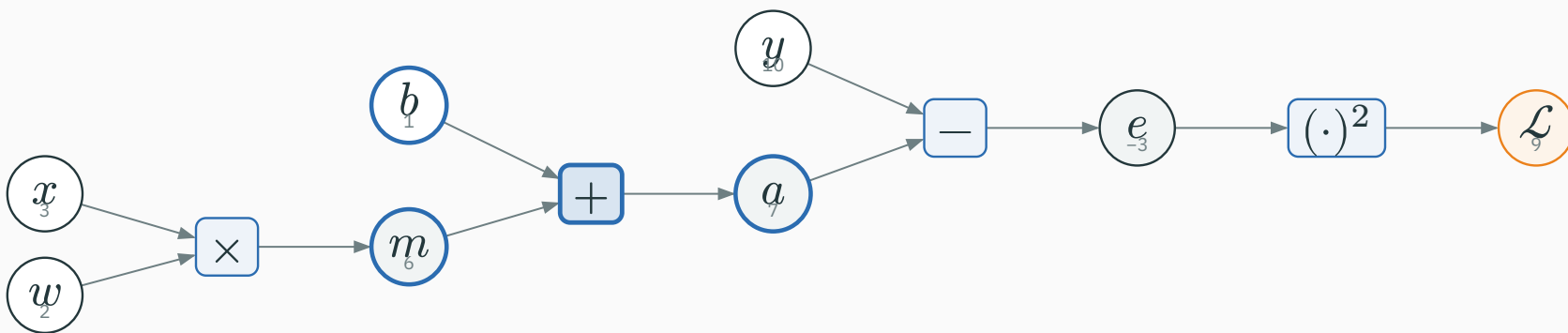
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The target y also receives a derivative. We would use $\partial \mathcal{L} / \partial y$ if y were itself a learned quantity.

Addition operation $a = m + b$ **copies** its gradient:

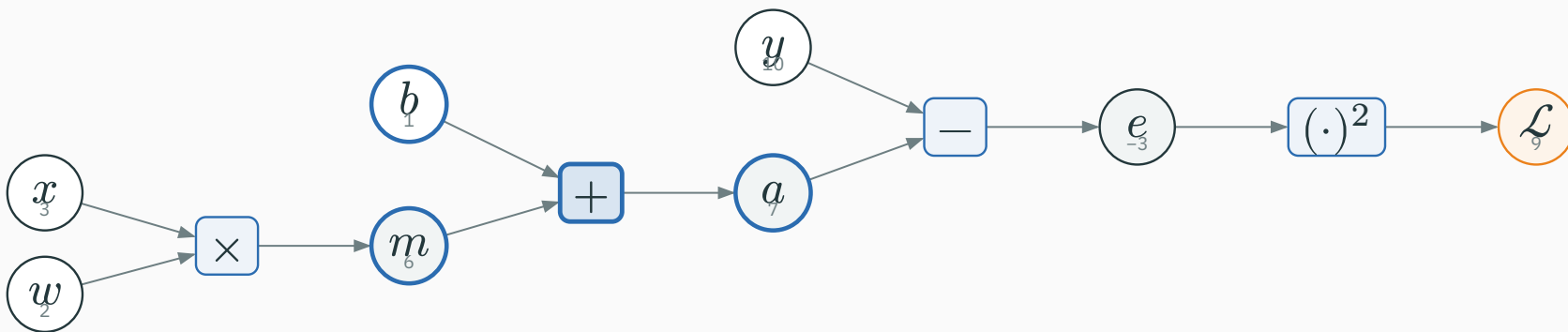
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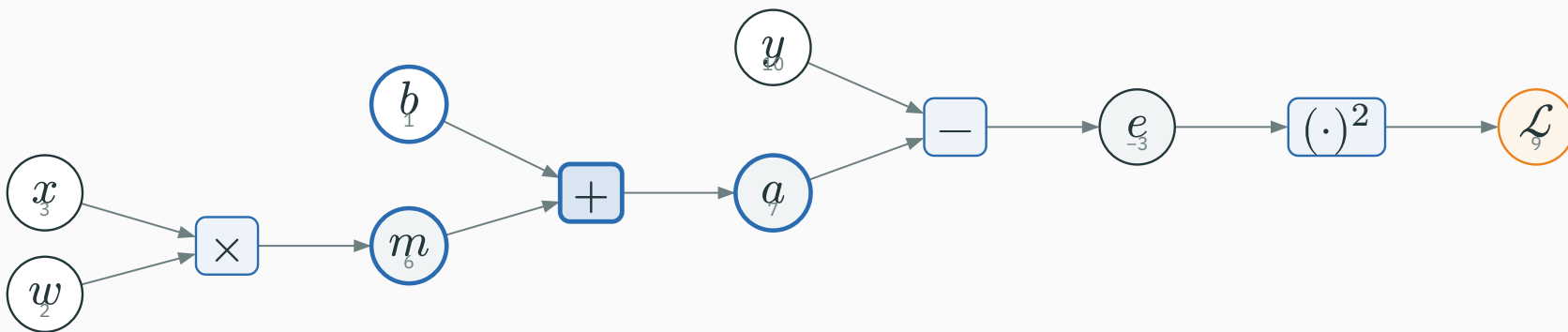
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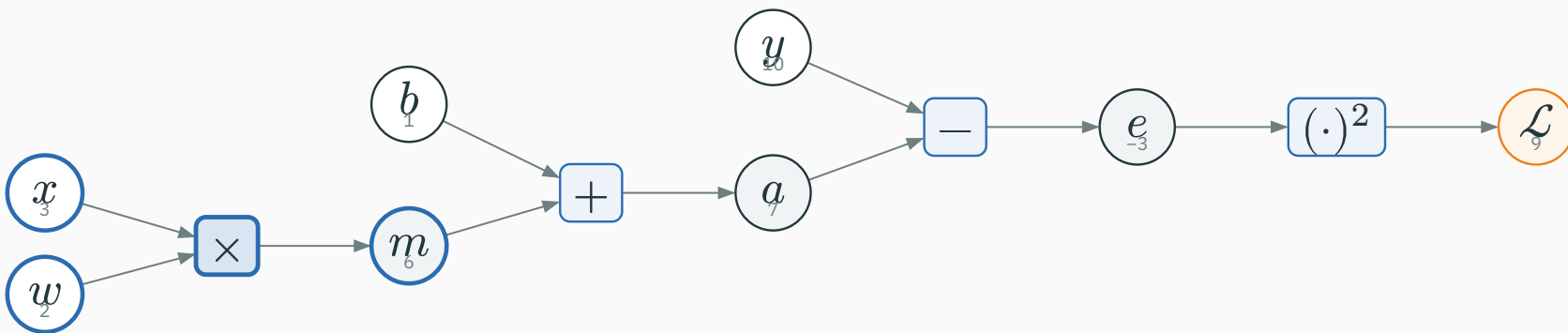
$$\frac{\partial \mathcal{L}}{\partial m} = \frac{\partial \mathcal{L}}{\partial a} = -6, \quad \frac{\partial \mathcal{L}}{\partial b} = -6.$$

So $\partial \mathcal{L} / \partial b = -6$ is one of our final answers.

Multiplication operation $m = wx$ **sends the other input:**

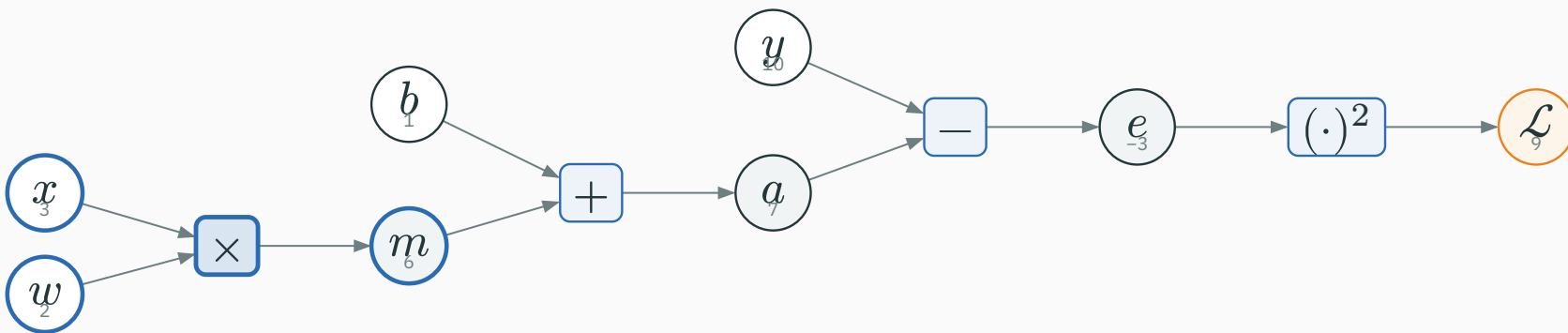
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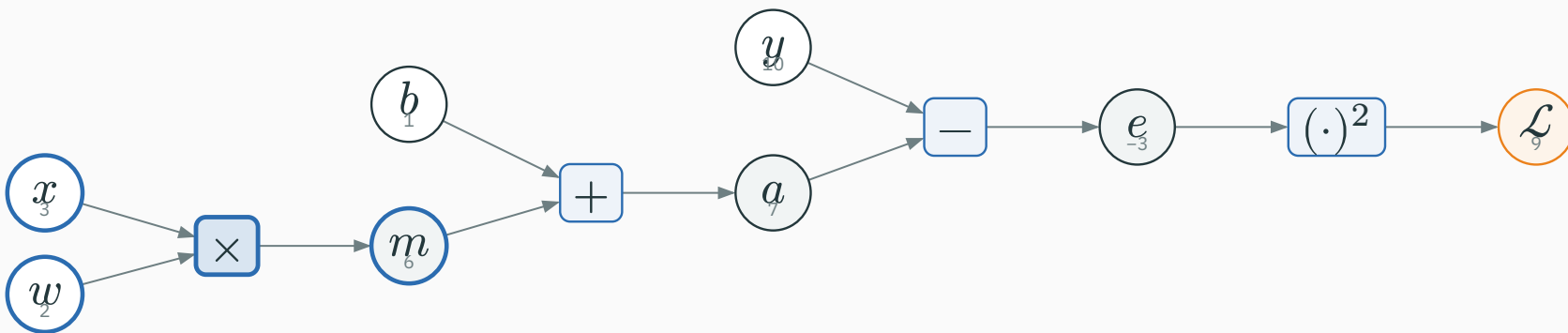
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$$\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial m} \cdot x = -6 \cdot 3 = -18$$

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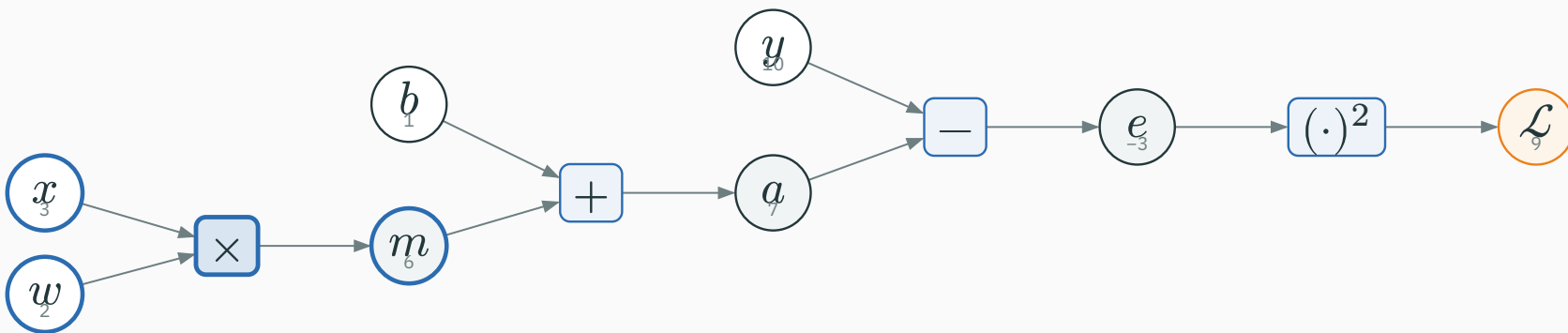


$$\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial m} \cdot x = -6 \cdot 3 = -18$$

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial m} \cdot w = -6 \cdot 2 = -12$$

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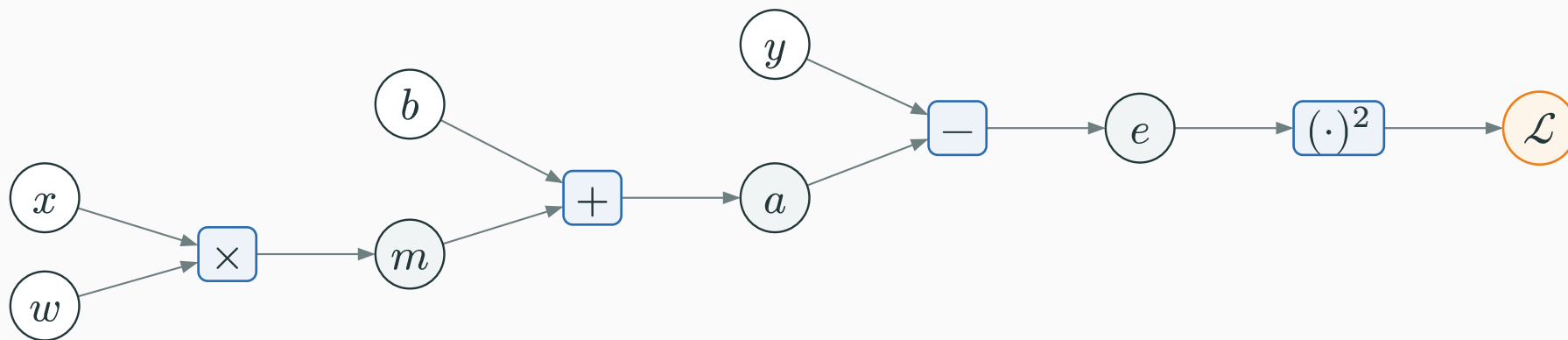
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One backward sweep produced **all four** partial derivatives — no formula re-derivation per parameter.

All stored values and their gradients



forward	$w = 2$	$x = 3$	$m = 6$	$b = 1$	$a = 7$	$y = 10$	$e = -3$	$\mathcal{L} = 9$
gradient	$g_w = -18$	$g_x = -12$	$g_m = -6$	$g_b = -6$	$g_a = -6$	$g_y = 6$	$g_e = -6$	$g_{\mathcal{L}} = 1$

$g_q \equiv \partial \mathcal{L} / \partial q$; operation boxes apply local rules, while stored values receive accumulated gradients

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Backprop and symbolic calculus give the same derivatives.

PyTorch executes the same graph

```
import torch

w = torch.tensor(2.0, requires_grad=True)
x = torch.tensor(3.0, requires_grad=True)
b = torch.tensor(1.0, requires_grad=True)
# Targets normally do not track gradients; enable it here to inspect dL/dy.
y = torch.tensor(10.0, requires_grad=True)

m = w * x
a = m + b
e = a - y
L = e**2
for value in (m, a, e, L):
    value.retain_grad()      # keep intermediate grads for inspection

L.backward()
print([t.item() for t in (m, a, e, L)])
# [6.0, 7.0, -3.0, 9.0]
print([t.grad.item() for t in (L, e, a, m, w, x, b, y)])
# [1.0, -6.0, -6.0, -6.0, -18.0, -12.0, -6.0, 6.0]
```

In our graph	In PyTorch	Meaning
stored scalar value	Tensor	one stored forward value
local backward rule	grad_fn	operation that produced a non-leaf tensor
saved forward value	saved by the operation	operand needed by a local derivative
reverse sweep	L.backward()	seed at $\mathcal{L}$; visit operations in reverse order
leaf derivative	p.grad	$\partial \mathcal{L} / \partial p$, accumulated with +=
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reverse mode = direction · backprop = algorithm · autograd = engine: record → save → replay

We found $\partial\mathcal{L}/\partial w = -18$. For a small positive learning rate, which way does gradient descent move w ?

A. Decrease w

B. Increase w

C. Leave w unchanged

D. The sign tells us nothing

Answer: a negative gradient means increase the parameter

A ANSWER

Correct: B — Increase w

Why: Gradient descent subtracts the gradient: $w \leftarrow w - \eta(-18) = w + 18\eta$.

A small gradient step reduces this loss

Take $\eta = 0.01$ and update:

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Backprop supplied the direction. The learning rate controls how far we move; a later optimization lecture studies that choice.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/autograd

Build a scalar computation graph, run the forward pass, then click **backward** to watch each $\partial \mathcal{L} / \partial u$ fill in — exactly the $(wx + b - y)^2$ example above.

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Trace which local rule changes when the final squared-error operation is replaced with $|e|$.

Practice the scalar graph three ways

I INTERACTIVE

See it move

Step through values and loss derivatives in the interactive graph ↗.

Build it by hand

Run the manual-backprop notebook ↓.

Let PyTorch build it

Run the autograd notebook ↓.

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Easy	change one input and recompute the forward pass
Medium	derive all four gradients before calling <code>backward()</code>
Hard	replace the square by $ e $; state what happens at $e = 0$

From scalars to vectors

We store gradients as column vectors. The name of the local object depends on the input and output shapes:

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Operation shape	Local object	Name
$u \in \mathbb{R} \rightarrow v \in \mathbb{R}$	$\partial v / \partial u \in \mathbb{R}$	derivative
$\mathbf{x} \in \mathbb{R}^d \rightarrow z \in \mathbb{R}$	$\partial z / \partial \mathbf{x} \in \mathbb{R}^d$	gradient
$\mathbf{x} \in \mathbb{R}^d \rightarrow \mathbf{z} \in \mathbb{R}^m$	$J_{ij} = \partial z_i / \partial x_j$	Jacobian $\mathbf{J} \in \mathbb{R}^{m \times d}$

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a Jacobian contains one local derivative for every output–input pair

A 2×2 Jacobian is four familiar derivatives

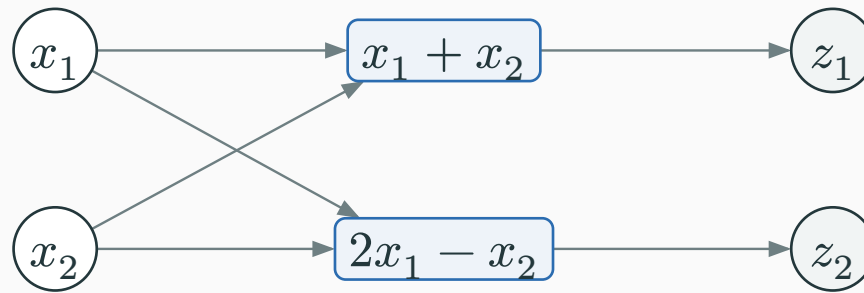
Take a vector operation with two inputs and two outputs:

$$z_1 = x_1 + x_2, \quad z_2 = 2x_1 - x_2.$$

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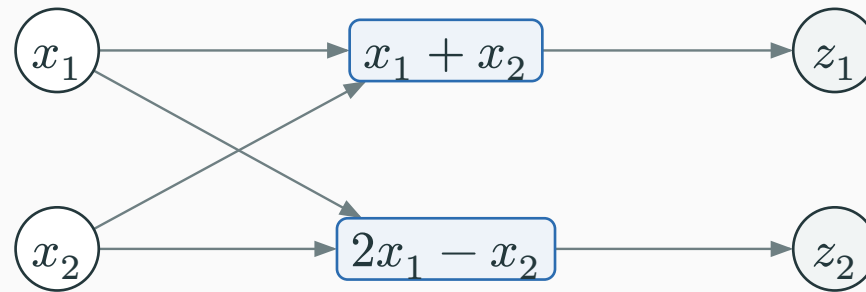
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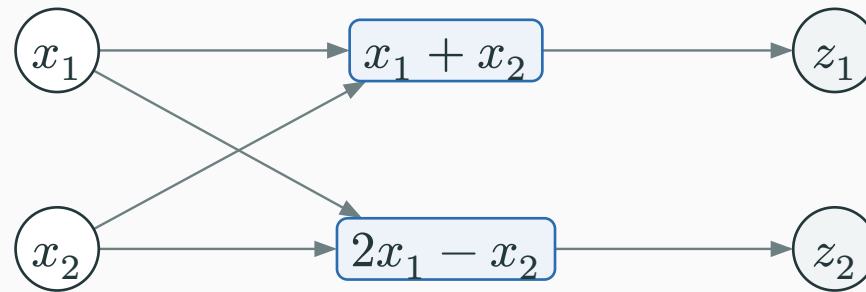


Output	Input x_1	Input x_2
z_1	$\partial z_1 / \partial x_1 = 1$	$\partial z_1 / \partial x_2 = 1$
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A 2×2 Jacobian is four familiar derivatives

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$$\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \cdot \text{rows are outputs} \cdot \text{columns are inputs}$$

Vector chain rule: make the shapes fit

For $x \in \mathbb{R}^d \rightarrow z \in \mathbb{R}^m$, let the **local Jacobian** be

$$J_{ij} = \partial z_i / \partial x_j, \quad J \in \mathbb{R}^{m \times d}.$$

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| When $d = m = 1$, J is one number and this is the scalar chain rule from earlier.

The transpose adds every output's contribution

For the previous operation, suppose the arriving gradient is

$$\mathbf{g}_z = \partial \mathcal{L} / \partial \mathbf{z} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

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$\partial \mathcal{L} / \partial x_1$	$3(1) + 4(2) = 11$
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the vector rule is the scalar branch rule applied to every input

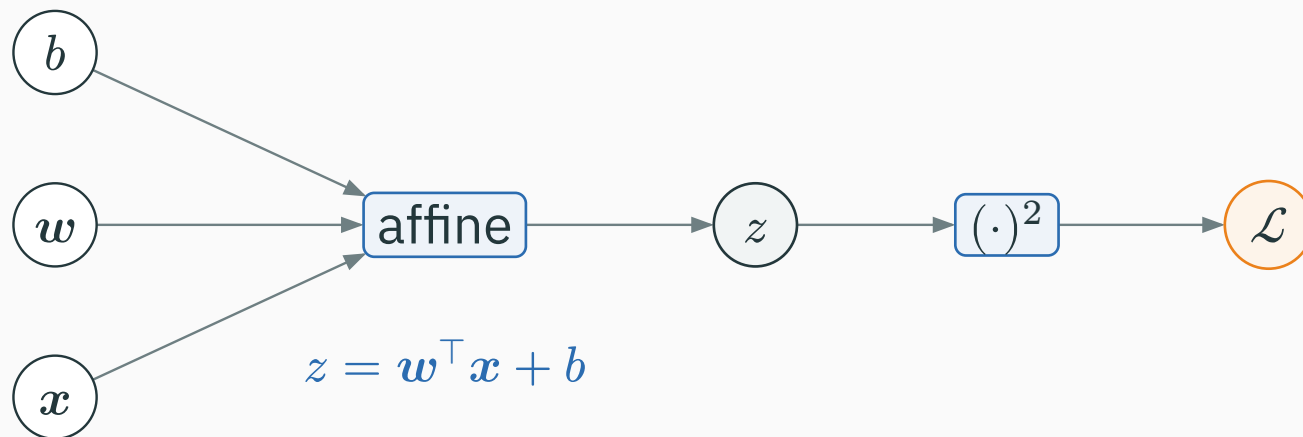
Affine means a weighted sum plus a bias:

$$z = \mathbf{w}^\top \mathbf{x} + b = \sum_{j=1}^d w_j x_j + b.$$

An affine operation adds scalar products and a bias

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Change one scalar while holding every other value fixed

Start from $z = \sum_j w_j x_j + b$. Change only one input by a small amount Δ :

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Change	Resulting change in z	Divide by Δ
$w_i \rightarrow w_i + \Delta$	$((w_i + \Delta)x_i) - w_i x_i = x_i \Delta$	$\partial z / \partial w_i = x_i$
$x_i \rightarrow x_i + \Delta$	$w_i(x_i + \Delta) - w_i x_i = w_i \Delta$	$\partial z / \partial x_i = w_i$
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local derivative = change in the operation's output $\div$ change in one input

Stack the coordinate derivatives into gradient vectors

The derivatives with respect to the coordinates of a vector are stored in one column:

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the vector formulas are just the scalar coordinate derivatives stacked together

Affine forward: multiply componentwise, then add

Use $x = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$, $w = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, and $b = 0$.

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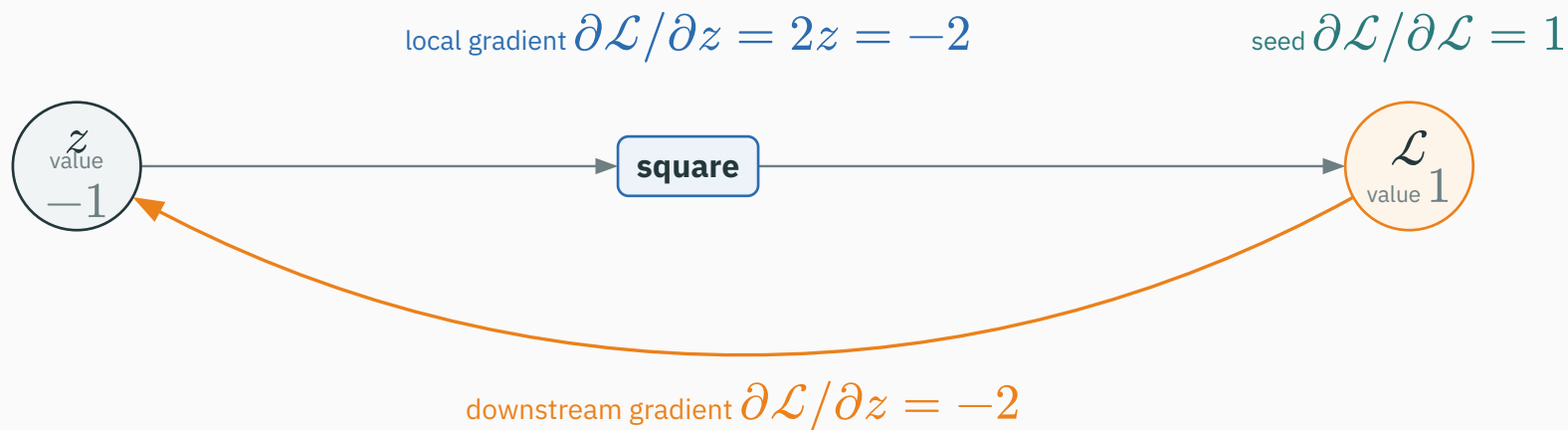
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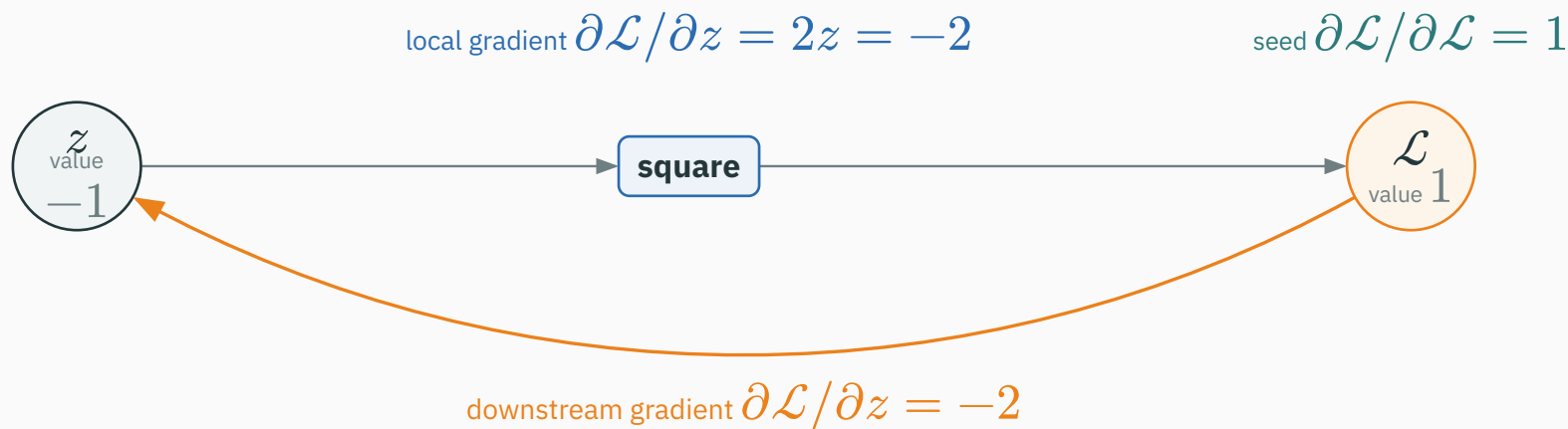
$$\mathcal{L} = z^2 = (-1)^2 = 1.$$

dot product = multiply matching entries, then sum

First backward step: the square operation



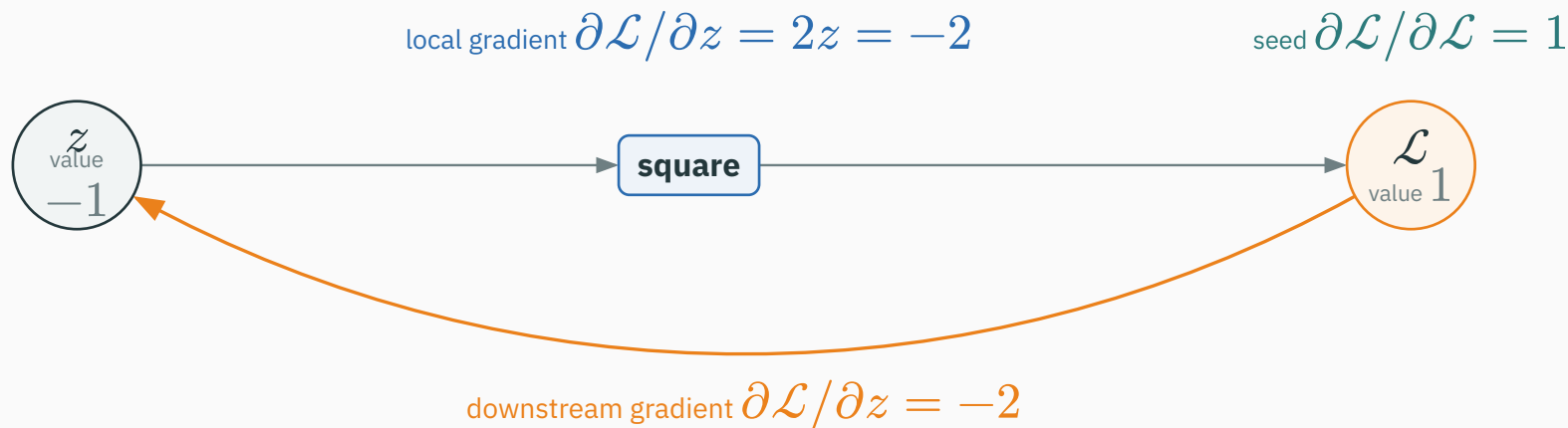
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Seed $\partial \mathcal{L} / \partial \mathcal{L} = 1$. The square has local derivative $2z = -2$, so

$$\frac{\partial \mathcal{L}}{\partial z} = 1 \cdot 2z = -2.$$

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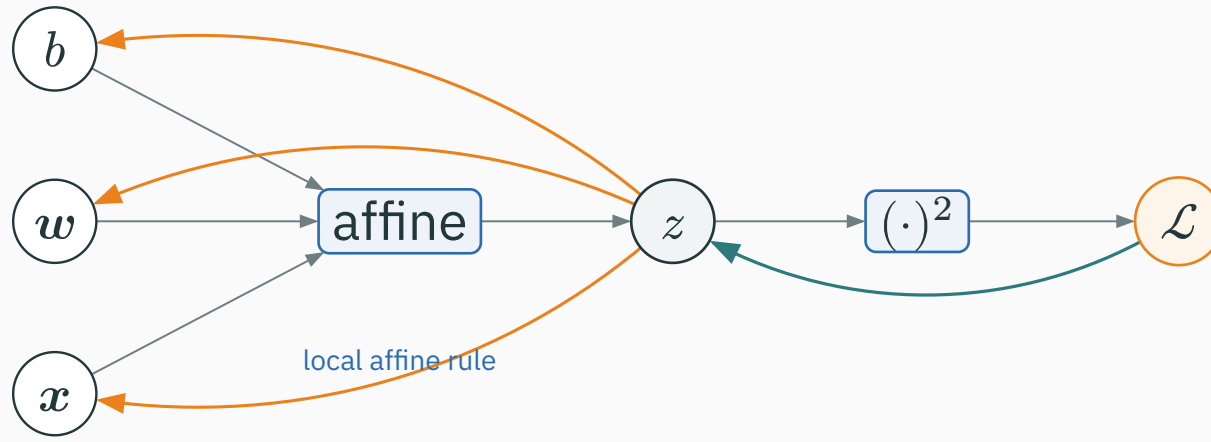


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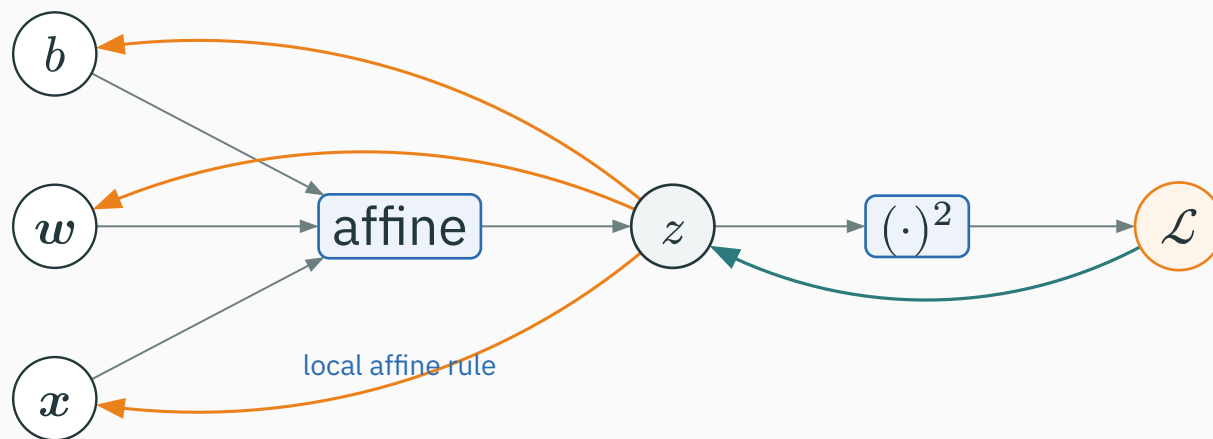
$$\frac{\partial \mathcal{L}}{\partial z} = 1 \cdot 2z = -2.$$

the square returns -2 ; it now arrives at the affine operation

Affine backward: apply the three local gradients

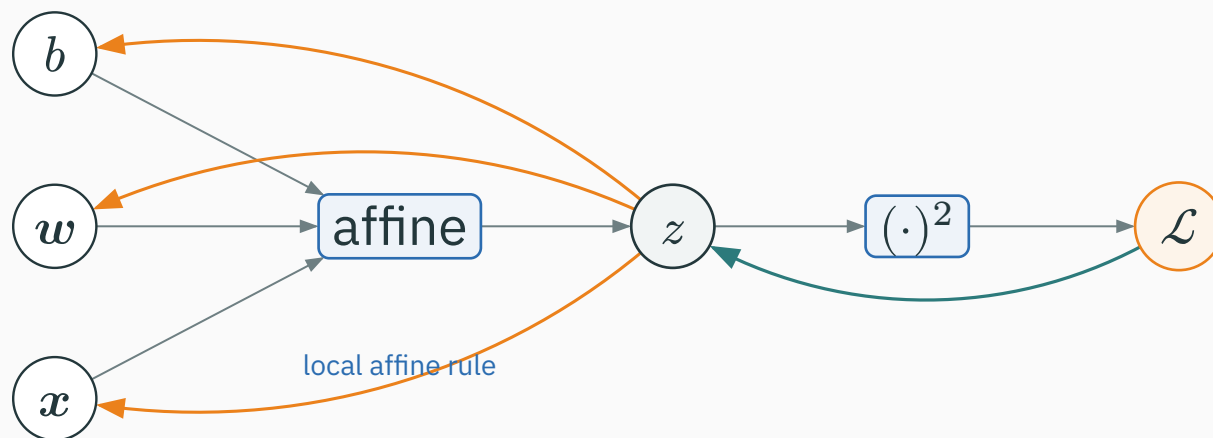


Affine backward: apply the three local gradients



Let $g_z = \partial \mathcal{L} / \partial z = -2$ arrive at the affine operation. Its **local gradients** send back:

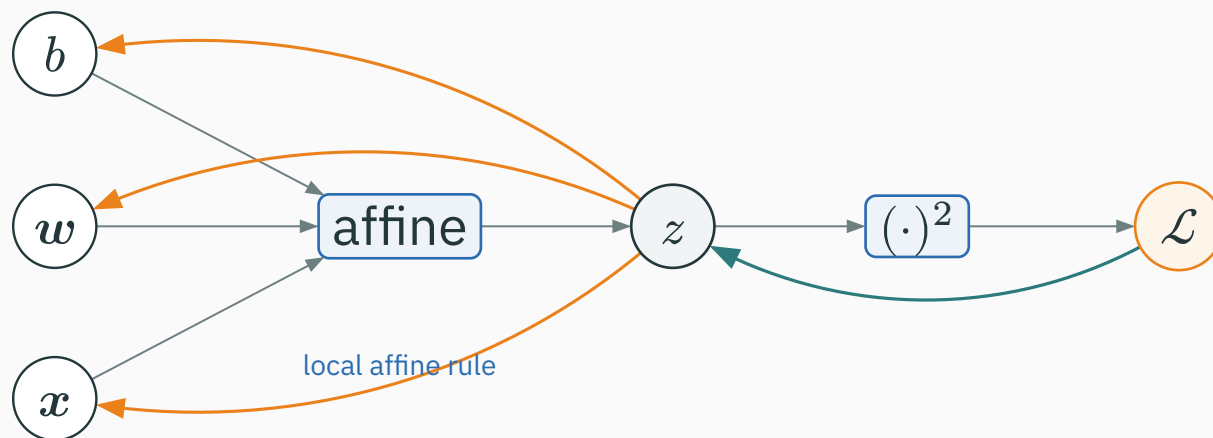
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Input	Local gradient	Downstream gradient
w	$\partial z / \partial w = x$	$\partial \mathcal{L} / \partial w = (-2)x = (-4, 2)^\top$
x	$\partial z / \partial x = w$	$\partial \mathcal{L} / \partial x = (-2)w = (-2, -6)^\top$
b	$\partial z / \partial b = 1$	$\partial \mathcal{L} / \partial b = -2$

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one arriving gradient g_z produces three separate downstream gradients

PyTorch gives the same vector gradients

```
import torch

x = torch.tensor([2., -1.], requires_grad=True)
w = torch.tensor([1., 3.], requires_grad=True)
b = torch.tensor(0., requires_grad=True)

z = w @ x + b
L = z**2
L.backward()

print(z.item(), L.item())  # -1.0, 1.0
print(w.grad)              # tensor([-4., 2.])
print(x.grad, b.grad)     # tensor([-2., -6.]), tensor(-2.)
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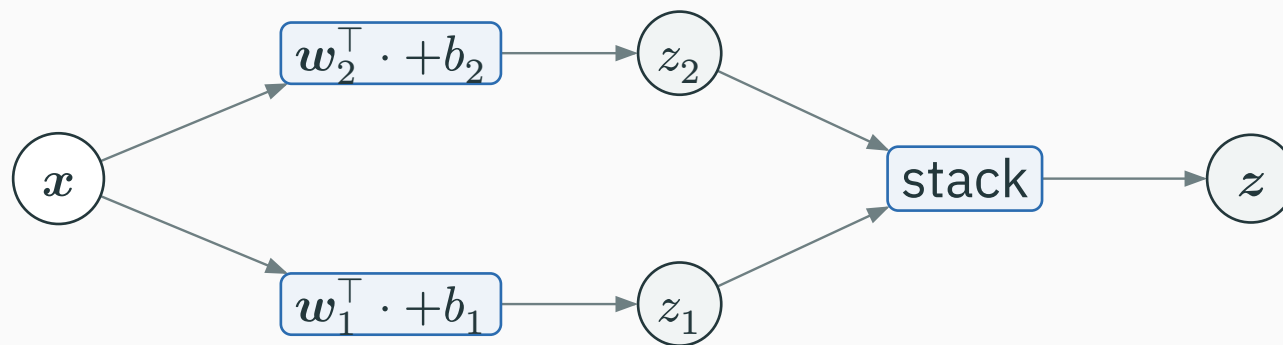
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print(x.grad, b.grad)     # tensor([-2., -6.]), tensor(-2.)
```

@ performs the dot product; backward() still applies the same local rules

From one scalar output to a vector output

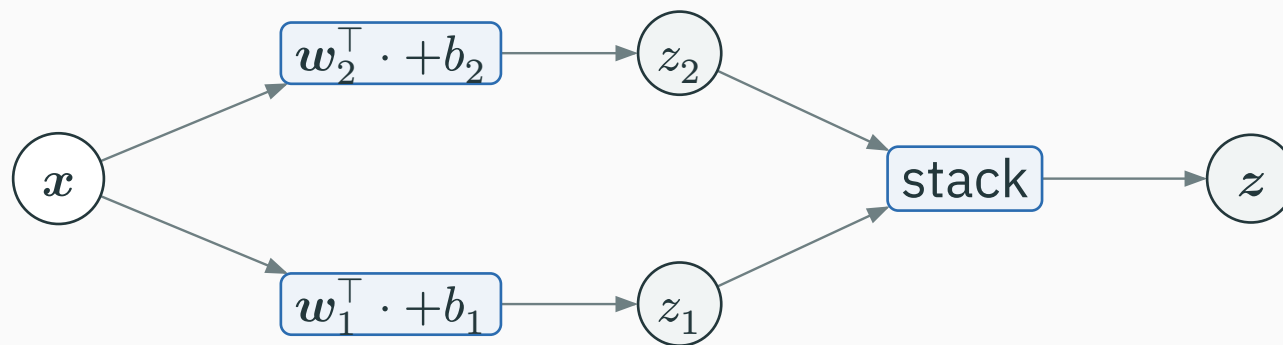
A dense layer repeats one familiar neuron

Every row of $\mathbf{W}$ is one affine neuron. All rows read the same input x .



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ROW 1 • neuron 1

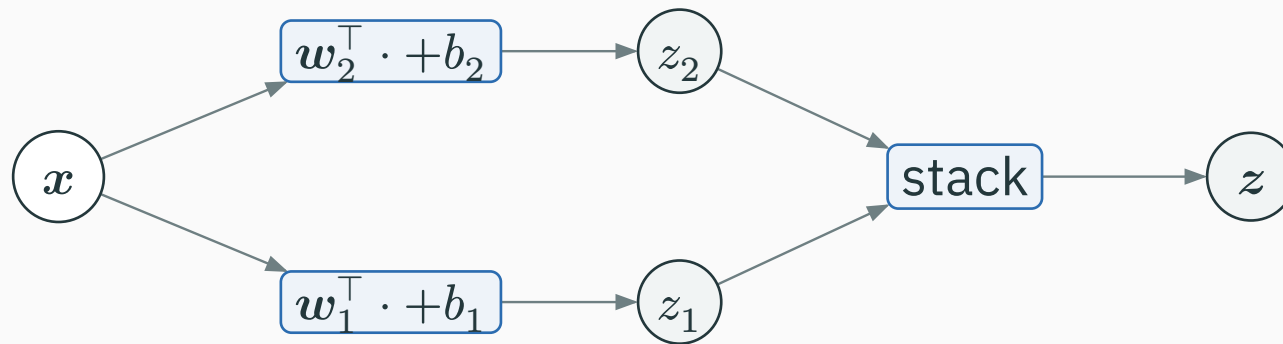
ROW 2 • neuron 2

$$z_1 = w_1^\top x + b_1$$

$$z_2 = w_2^\top x + b_2$$

A dense layer repeats one familiar neuron

Every row of W is one affine neuron. All rows read the same input x .



ROW 1 · neuron 1

ROW 2 · neuron 2

$$z_1 = w_1^\top x + b_1$$

$$z_2 = w_2^\top x + b_2$$

stack the outputs: $z = Wx + b$

Forward, row 1: do one dot product

D DERIVATION

$$\boldsymbol{x} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \boldsymbol{w}_1^\top = (1, 3), \quad b_1 = 0$$

Forward, row 1: do one dot product

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$$z_1 = \underbrace{1(2)}_{\text{first input}} + \underbrace{3(-1)}_{\text{second input}} + 0 = -1$$

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| Nothing new happened: multiply matching entries, add them, then add the bias.

Forward, row 2: repeat, then stack

D DERIVATION

$$\mathbf{w}_2^\top = (-2, 1), \quad b_2 = 1$$

Forward, row 2: repeat, then stack

D DERIVATION

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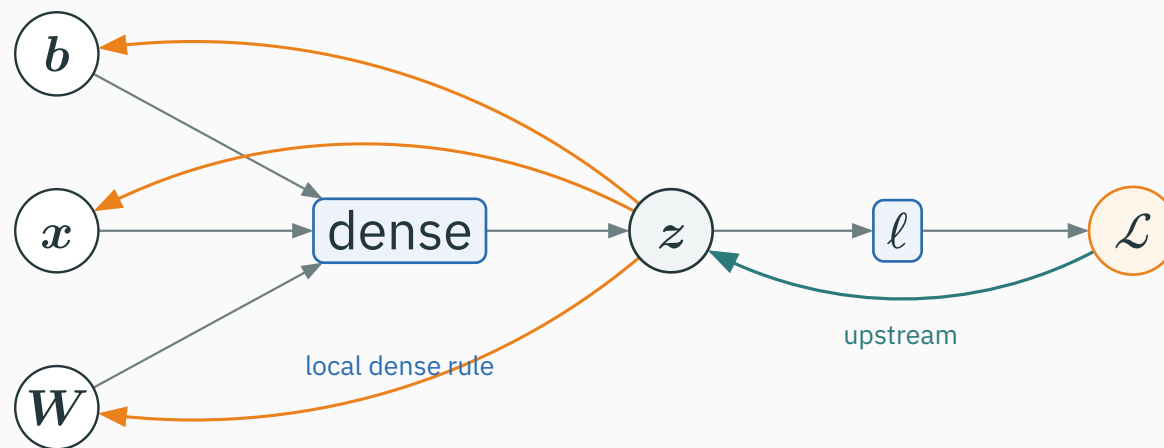
$$z_2 = (-2)(2) + 1(-1) + 1 = -4$$

$$z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} -1 \\ -4 \end{pmatrix}$$

$$W = \begin{pmatrix} 1 & 3 \\ -2 & 1 \end{pmatrix} \text{ simply stores the two neuron rows together}$$

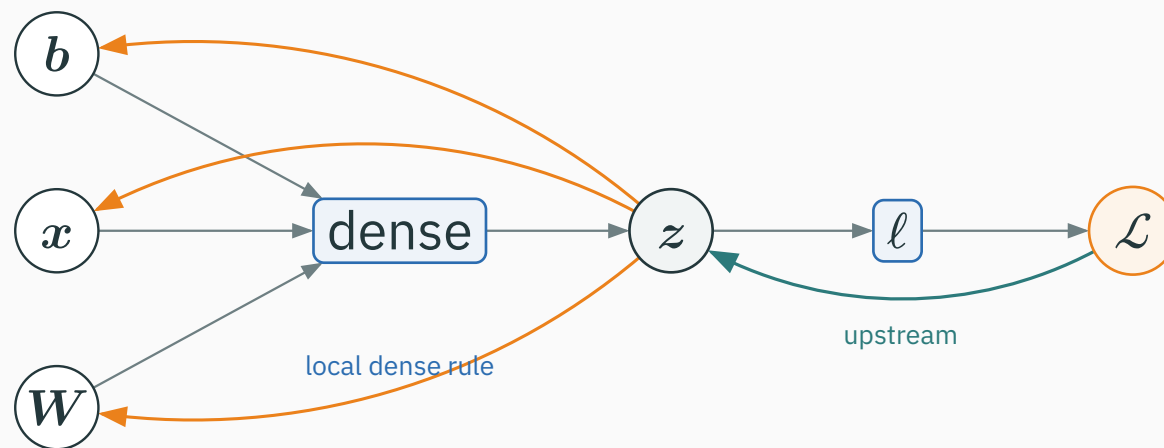
Backward starts with one gradient per output

Suppose the later loss sends $g_z = \partial \mathcal{L} / \partial z = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ back to this layer.



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TO INPUT

TO WEIGHTS

TO BIAS

$$g_x = W^\top g_z$$

$$g_W = g_z x^\top$$

$$g_b = g_z$$

Input gradient: both outputs contribute and add

D DERIVATION

Each input coordinate was used by both output neurons, so two paths return to it.

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x_1 receives

$$4(1) + (-2)(-2) = 8$$

from row 1 + from row 2

x_2 receives

$$4(3) + (-2)(1) = 10$$

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the matrix product performs the same branch addition for every input

Weight gradient: one outer product fills the matrix

D DERIVATION

Output i sends $g_i x_j$ to weight $W_{i,j}$. Writing every pair at once gives an outer product.

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$$\mathbf{g}_W = \begin{pmatrix} 4 \\ -2 \end{pmatrix} \begin{pmatrix} 2 & -1 \end{pmatrix} = \begin{pmatrix} 8 & -4 \\ -4 & 2 \end{pmatrix}$$

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ROWS FOLLOW OUTPUTS

row 1 uses $g_1 = 4$; row 2 uses
 $g_2 = -2$

BIAS HAS LOCAL DERIVATIVE 1

$$\mathbf{g}_b = \mathbf{g}_z = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$

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BIAS HAS LOCAL DERIVATIVE 1

$$\mathbf{g}_b = \mathbf{g}_z = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$

$$\text{shapes: } \mathbf{g}_W \in \mathbb{R}^{2 \times 2} \cdot \mathbf{g}_b \in \mathbb{R}^2 \cdot \mathbf{g}_x \in \mathbb{R}^2$$

A batch reuses the same $\mathbf{W}, \mathbf{b}$ three times

For each row $\mathbf{x}^i$, compute $\mathbf{z}^i = \mathbf{W}\mathbf{x}^i + \mathbf{b}$ and residual $\mathbf{r}^i = \mathbf{z}^i - \mathbf{y}^i$.

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EXAMPLE 1

$$x = (2, -1)$$

$$y = (-5, -2)$$

$$z = (-1, -4)$$

$$r = (4, -2)$$

EXAMPLE 2

$$x = (-1, 2)$$

$$y = (7, 1)$$

$$z = (5, 5)$$

$$r = (-2, 4)$$

EXAMPLE 3

$$x = (2, 2)$$

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$$\mathcal{L} = \frac{1}{3} \sum_{i=1}^3 \frac{1}{2} \|r^i\|^2 = \frac{10+10+1}{3} = 7$$

A batch reuses the same W, b three times

For each row x^i , compute $z^i = Wx^i + b$ and residual $r^i = z^i - y^i$.

EXAMPLE 1

$$\begin{aligned}x &= (2, -1) \\y &= (-5, -2) \\z &= (-1, -4) \\r &= (4, -2)\end{aligned}$$

EXAMPLE 2

$$\begin{aligned}x &= (-1, 2) \\y &= (7, 1) \\z &= (5, 5) \\r &= (-2, 4)\end{aligned}$$

EXAMPLE 3

$$\begin{aligned}x &= (2, 2) \\y &= (7, -2) \\z &= (8, -1) \\r &= (1, 1)\end{aligned}$$

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one parameter tensor, three forward paths, one mean loss

Batch backward: average the three contributions

Each example proposes an unscaled weight contribution $G^i = r^i (x^i)^\top$.

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EXAMPLE 1

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EXAMPLE 2

$$G^2 = \begin{pmatrix} 2 & -4 \\ -4 & 8 \end{pmatrix}$$

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$$g_W = \frac{1}{3}(G^1 + G^2 + G^3) = \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$$

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Mean reduction averages example contributions. A **sum** reduction would be exactly three times larger.

PyTorch verifies the single example and the batch

```
X = torch.tensor([[ 2., -1.], [-1., 2.], [2., 2.]])
Y = torch.tensor([[-5., -2.], [ 7., 1.], [7., -2.]])
W = torch.tensor([[1., 3.], [-2., 1.]], requires_grad=True)
b = torch.tensor([0., 1.], requires_grad=True)

Z = X @ W.T + b
loss = 0.5 * ((Z - Y)**2).sum(dim=1).mean()
loss.backward()

print(loss.item()) # 7.0
print(W.grad)      # [[ 4., -2.], [-2.,  4.]]
print(b.grad)      # [1., 1.]
```

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I INTERACTIVE

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Run the Dense + Batch Colab ↗ to reveal every example's outer product and verify that three correctly scaled micro-batches equal the full-batch gradient.

The training loop now has a concrete batch

I INTERACTIVE

```
for X, Y in loader:
    optimizer.zero_grad()      # clear the previous update's buffers
    Z = model(X)               # all examples reuse the same parameters
    loss = loss_fn(Z, Y)       # reduce example losses to one scalar
    loss.backward()            # aggregate this batch's contributions
    optimizer.step()           # update once
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<code>model(X)</code>	records one graph whose rows share the same parameters
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clear → batch forward → scalar loss → batch backward → update

Accumulate within an update; clear between updates

Situation	What accumulates	Correct boundary
three examples in one mean loss	three path contributions in one backward pass	divide by $B = 3$ through the loss reduction
three micro-batches for one update	three scaled <code>.backward()</code> calls in <code>.grad</code>	clear once before the first; step once after the third
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Branch addition, batch aggregation, and `.grad += ...` share the same mathematics—but the intended update window determines when to clear.

A tiny neural network

ReLU is what prevents two dense layers from collapsing into one

Without a nonlinearity, two affine maps are still one affine map:

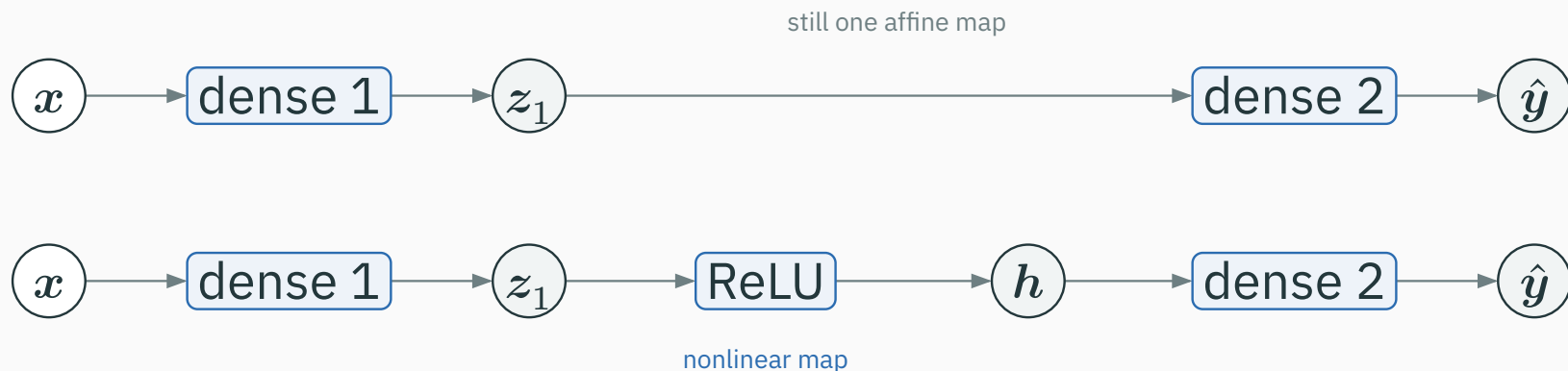
$$\begin{aligned} \mathbf{W}_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2 \\ = (\mathbf{W}_2\mathbf{W}_1)\mathbf{x} + (\mathbf{W}_2\mathbf{b}_1 + \mathbf{b}_2). \end{aligned}$$

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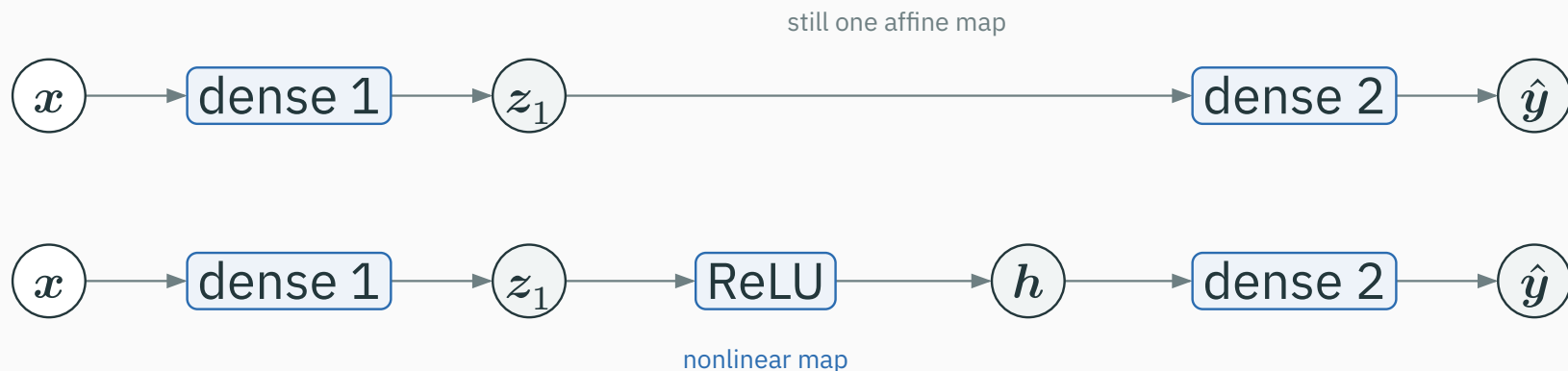


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ReLU makes the composition nonlinear; depth can now represent a richer function

A tiny MLP composes three vector blocks

The network now repeats the operations we already know: affine map, elementwise ReLU, affine map, scalar loss.

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$$z_1 = W_1 x + b_1$$

$$h = \text{ReLU}(z_1)$$

$$\hat{y} = W_2 h + b_2$$

$$\mathcal{L} = \sum_i (\hat{y}_i - y_i)^2$$

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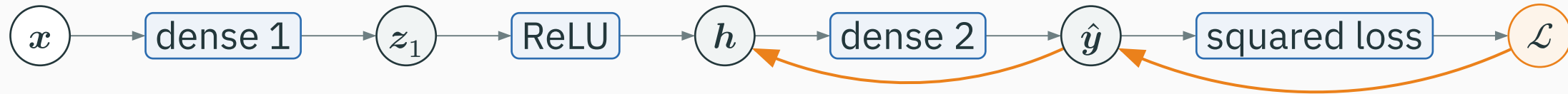
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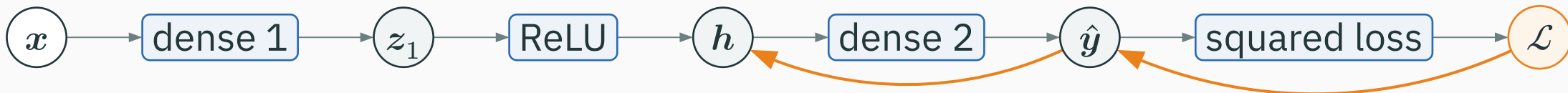
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dense → ReLU → dense → squared loss

Backward, part 1: loss $\rightarrow$ dense 2 $\rightarrow$ hidden activations

D DERIVATION

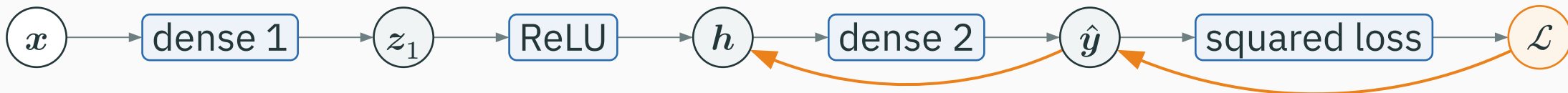




1. Squared loss. Start with the seed $\partial \mathcal{L} / \partial \mathcal{L} = 1$. The loss operation returns

$$g_{\hat{y}} = 2(\hat{y} - y).$$

This says how increasing each prediction coordinate would change the loss.



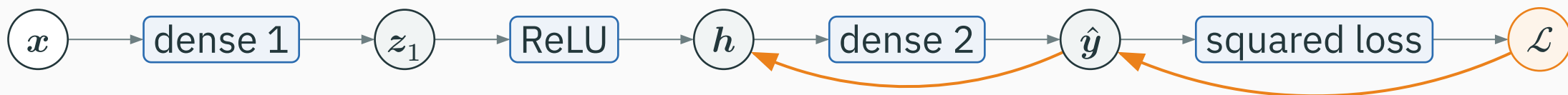
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Second dense layer. It receives $g_{\hat{y}}$ and returns a gradient to its input h :

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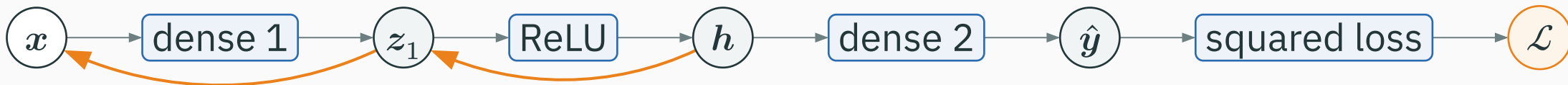
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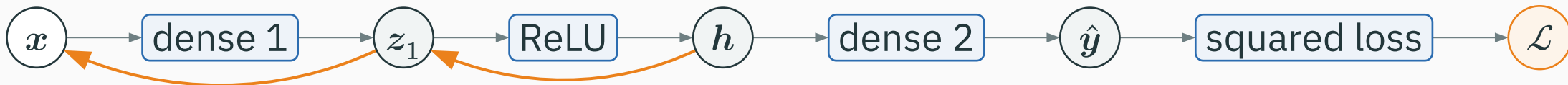
$$g_h = W_2^\top g_{\hat{y}}.$$

the returned g_h is now the arriving gradient at ReLU

Backward, part 2: ReLU $\rightarrow$ dense 1 $\rightarrow$ input

D DERIVATION

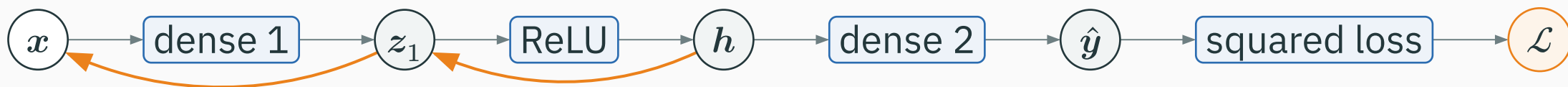




3. ReLU. Apply its local gate coordinate by coordinate:

$$g_{z_1} = g_h \odot \mathbb{1}[z_1 > 0].$$

An active unit passes its arriving gradient; an inactive unit returns zero.

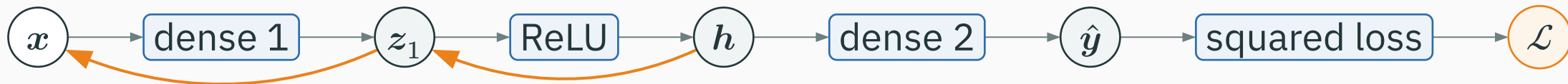


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each returned gradient becomes the arriving gradient for the block immediately to its left

The same tiny MLP in PyTorch

```
import torch
from torch import nn

model = nn.Sequential(
    nn.Linear(2, 3), nn.ReLU(), nn.Linear(3, 2)
)
x = torch.tensor([2., -1.])
y = torch.tensor([1., 0.])

y_hat = model(x)
loss = ((y_hat - y)**2).sum()
loss.backward()

for name, p in model.named_parameters():
    print(name, tuple(p.shape), tuple(p.grad.shape))
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PyTorch records the three blocks and applies their local rules in reverse

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/autograd

Step through the forward values, then follow one arriving gradient through a dense operation, a ReLU, and the preceding dense operation.

INTERACTIVE

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For practice, predict each gradient's shape before revealing its value in the interactive.

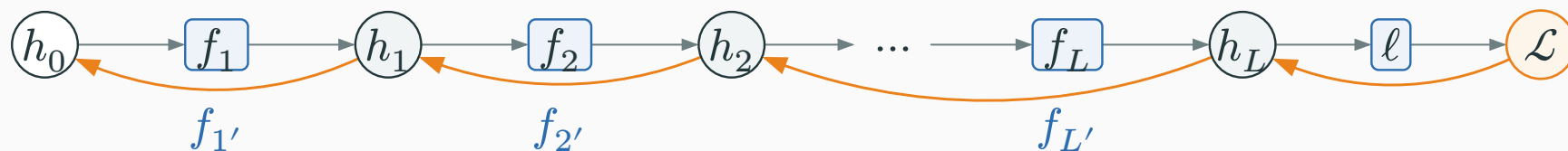
When gradients cross many layers

Deep networks repeat the same multiplication

For a scalar chain $h_0 \rightarrow h_1 \rightarrow \dots \rightarrow h_L \rightarrow \mathcal{L}$:

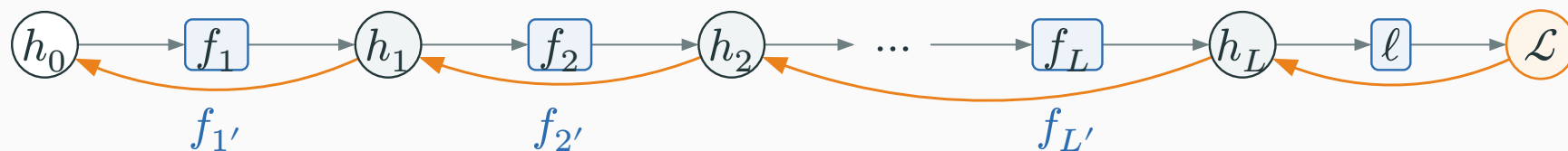
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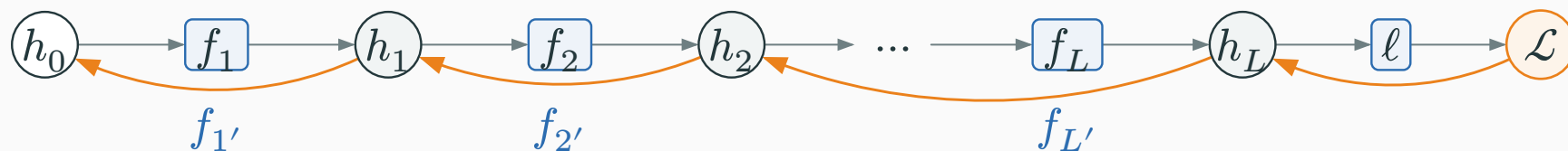
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$$\frac{\partial \mathcal{L}}{\partial h_0} = \frac{\partial \mathcal{L}}{\partial h_L} \cdot f_{L'}(h_{L-1}) \dots f_{1'}(h_0).$$

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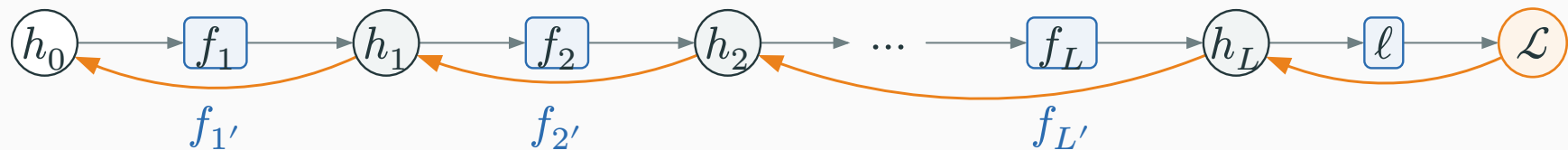


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depth creates a product of many local slopes

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depth creates a product of many local slopes

Vector layers do the same operation with matrix–vector products. The intuition is unchanged: repeatedly small factors shrink a gradient; repeatedly large factors grow it.

Ten ordinary-looking slopes can create three regimes

Start with an arriving loss derivative of 1 and multiply the same local slope ten times:

Ten ordinary-looking slopes can create three regimes

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Local slope	After 10 layers	Effect
0.5	$0.5^{10} \approx 0.001$	gradient nearly disappears
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1.5	$1.5^{10} \approx 57.7$	gradient grows rapidly

Ten ordinary-looking slopes can create three regimes

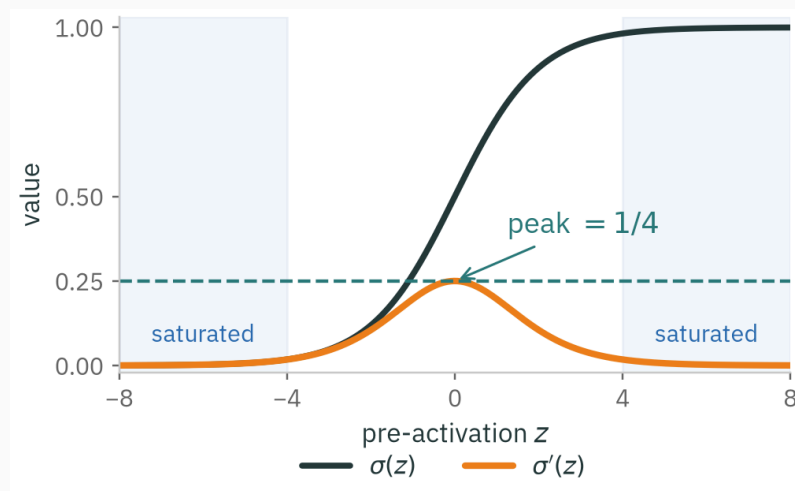
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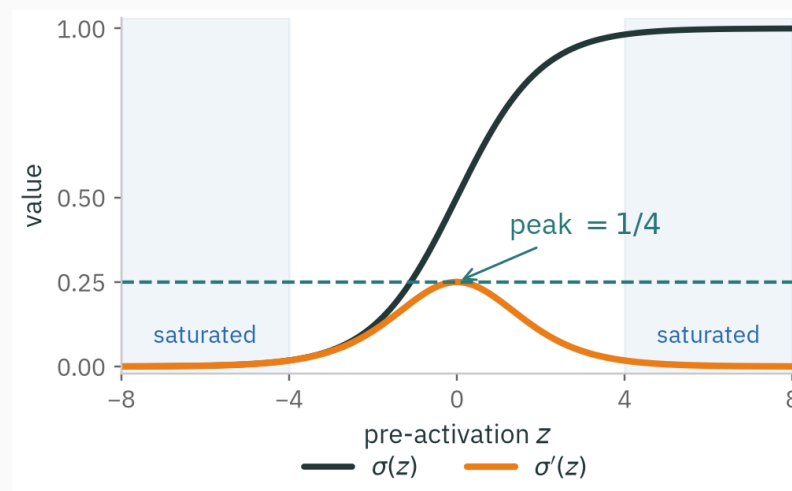
the chain rule explains both vanishing and exploding gradients

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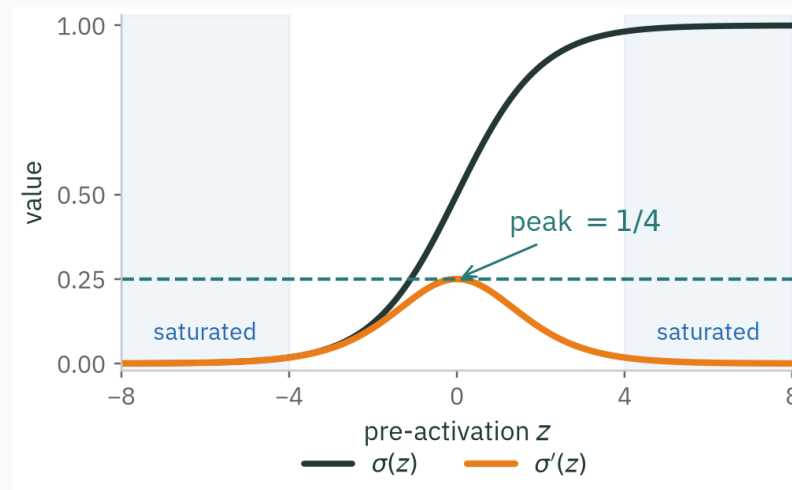


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each sigmoid layer multiplies the gradient by at most $1/4$

Repeated sigmoid derivatives can shrink the signal quickly. ReLU has slope 1 on active units and 0 on inactive units; weights and normalization also shape the full Jacobian.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/vanishing-gradients

Sweep depth and activation function; watch the gradient magnitude at layer 0 collapse for sigmoid and survive for ReLU.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/vanishing-gradients

Sweep depth and activation function; watch the gradient magnitude at layer 0 collapse for sigmoid and survive for ReLU.

Sweep depth to see repeated small local slopes rapidly erase an early-layer gradient.

Summary

Backprop in one sentence

Seed $\partial \mathcal{L} / \partial \mathcal{L} = 1$, then walk the graph **backward**, at each operation computing

downstream gradient = **upstream gradient** $\times$ **local derivative**,

and **adding** gradients wherever a variable branched.

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and **adding** gradients wherever a variable branched.

For vector operations, the **local derivative** can be a vector or matrix; the same color-coded chain rule applies.

Setting	Backward rule
scalar operation	$\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial v} \cdot \partial v / \partial u$
dense layer	$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}} \mathbf{x}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{x}} = \mathbf{W}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{z}}$
activation	$\frac{\partial \mathcal{L}}{\partial \mathbf{z}} = \frac{\partial \mathcal{L}}{\partial \mathbf{a}} \odot \varphi'(\mathbf{z})$
branch point	gradients add

We can compute $\partial \mathcal{L} / \partial \theta_j$ for every parameter.

Next: turn those gradients into reliable parameter updates.

Lecture 5 · Optimization for Deep Learning